

CAPSTONE PROJECT REPORT

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T.O.M.M.Y.: A Unified Multimodal Framework for Real-Time OS Control Using MediaPipe, Gaze Estimation, and Speech Recognition

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DECLARATION STATEMENT

We hereby declare that the research work reported in the dissertation/dissertation proposal entitled "**T.O.M.M.Y.: A Unified Multimodal Framework for Real-Time OS Control using MediaPipe, Gaze Estimation, and Speech Recognition**" in partial fulfillment of the requirement for the award of Degree for Bachelor of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under the supervision of our research supervisor Kaushal Pathak. We have not submitted this work elsewhere for any degree or diploma.

We understand that the work presented herewith is in direct compliance with Lovely Professional University's policy on plagiarism, intellectual property rights, and the highest standards of moral and ethical conduct. Therefore, to the best of our knowledge, the content of this dissertation represents an authentic and honest research effort conducted, in its entirety, by us. We are fully responsible for the contents of our dissertation work.

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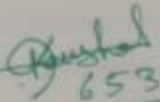
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CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Internship Project (CSE461) under my guidance and supervision. The present work is the result of their original investigation, effort, and study. No part of the work has ever been submitted for any other degree at any University. The Internship Project is fit for the submission and partial fulfilment of the conditions for the award of B.Tech degree in School of Computer Science and Engineering from Lovely Professional University, Phagwara.

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CHAPTER 1. INTRODUCTION

The development of Human-Computer Interaction (HCI) is an intriguing chronicle of software modifications and hardware limitations. The physical keyboard and mechanical (or optical) mouse have served as the main interface between human intent and digital execution for almost forty years. With the advent of Graphical User Interfaces (GUIs) in the 1980s, these paradigms were fundamentally solidified and have stayed relatively unchanged. The physical mechanics of dragging a piece of plastic across a desk to move a virtual arrow have not altered, despite the underlying computers' orders of magnitude increase in power.

It makes sense that the mouse and keyboard are so popular. They provide unmatched accuracy, instant tactile feedback, and complete dependability. Nevertheless, the user is forced into a rather constrictive position by this conventional paradigm. Both are necessary, hands to be physically tethered to a desk, mandates close physical proximity to the machine, and introduces significant ergonomic strain—evidenced by the rising rates of Carpal Tunnel Syndrome and repetitive strain injuries among modern professionals. More critically, the traditional hardware paradigm erects a massive barrier to entry for individuals with physical disabilities or limited motor control.

As computing transitions from stationary desktop environments to ubiquitous, integrated spatial environments, the way we control our operating systems must evolve. There is a very real push toward hands-free interaction, often branded as "spatial computing." Companies are attempting to solve this via expensive, wearable hardware like VR headsets or specialized infrared tracking rigs. However, requiring users to purchase thousands of dollars of proprietary hardware defeats the purpose of accessible computing.

Simultaneously, the software landscape has seen an explosion in the capabilities of artificial intelligence. Voice assistants are now capable of transcribing speech in real-time, and computer vision frameworks can track human biometrics with millimetre precision. Yet, when these technologies are applied to desktop operating systems, they usually fail to provide a cohesive experience. Current voice assistants excel at retrieving information ("What is the weather?") but fail completely at granular GUI control ("Drag the third file into the archive folder"). Current computer vision

frameworks can track a hand, but trying to use them as a persistent mouse replacement usually results in a shaky, unstable cursor that frustrates the user.

This research introduces T.O.M.M.Y. (Transformative Ocular and Multimodal Matrix Yield). T.O.M.M.Y. is not just another wrapper for an AI model. It is a fundamental rethinking of how machine learning algorithms are orchestrated at the operating system level. Instead of running a monolithic script that chokes under the weight of simultaneous video and audio processing, T.O.M.M.Y. utilizes a highly decoupled, multi-process architecture based on Inter-Process Communication (IPC). By isolating the computer vision pipeline and the natural language processing (NLP) engine into entirely separate, headless daemons, the framework circumvents the traditional bottlenecks of the Python programming language.

The ultimate goal of this thesis is to prove that high-fidelity, zero-latency, hands-free operating system control can be achieved using nothing more than a standard RGB webcam and a basic microphone, provided the underlying software architecture is correctly optimized for concurrent multimodal data streams.

CHAPTER 2. PROBLEM STATEMENT

2.1 PROFILE OF THE PROBLEM

The core problem driving this research is fundamentally split into two distinct profiles: ergonomic limitations on the human side, and severe architectural bottlenecks on the software side.

The Human Profile: It takes fine motor abilities to navigate a modern desktop operating system. The conventional mouse is sufficient for most users. However, the physical act of holding a mouse and pressing a mechanical switch to click is either painful or impossible for those with neuro-motor difficulties, repetitive strain injuries (such as carpal tunnel syndrome), or transient physical limitations. Additionally, users often have to interact with schematics or data on a screen while their hands are actively occupied or sterilized in professional settings like medical operating rooms, mechanic garages, or industrial fabrication labs. Workflow is severely disrupted in these situations by the physical keyboard and mouse. The issue is that existing solutions, such as simple voice dictation, do not have the spatial dimensions required to truly operate the machine;

The Software Profile: The technical issue is that high-level programming languages, especially Python, are very difficult to use for creating a real-time multimodal system. Since Python is the industry standard for machine learning, all of the top NLP and computer vision libraries are created for it. On the other hand, Python uses a Global Interpreter Lock (GIL). Python bytecodes cannot be executed simultaneously by multiple native threads thanks to the GIL.

In order to create a hands-free OS controller, developers usually implement a while loop that records a camera frame (cv2.VideoCapture), analyzes hand landmarks, and then records an audio segment (pyaudio) to process speech. The thread stalls as a result of both of these activities' significant I/O task stalling. The camera frame freezes if the microphone is active. Should

2.2. RATIONALE / SCOPE OF THE STUDY

T.O.M.M.Y. was inspired by these software annoyances. The capabilities of individual computer vision scripts and their combined performance as a persistent background service were shown to differ significantly. The goal is to show that by eschewing the monolithic script approach and putting in place a decentralized microservice architecture based on local networking sockets, we can achieve actual parallel execution in Python and totally eliminate the GIL constraint.

This study's scope is limited to deterministic control and local, offline execution.

Privacy and Latency Restrictions: The use of cloud-based APIs (such as Google Cloud Vision or Amazon Alexa) was specifically forbidden. For a system intended to operate continuously in the background, sending a continuous stream of a user's webcam feed and microphone audio to a remote server involves severe privacy infringement. Furthermore, the appearance of real-time control is totally destroyed by the round-trip delay of a network request, which is typically 200–500 milliseconds. As a result, only models that can operate on the "edge"—that is, directly on the user's local CPU—are included.

Deterministic Parsing over Generative AI: Furthermore, the NLP integration's scope specifically steers clear of generative AI (such as ChatGPT). Strict, predictable command execution is necessary for desktop operating systems. When a user says, "Close the browser," the system has to carry out the precise termination command. The possibility of "hallucinations"—in which the model chooses to type the word "browser" rather of performing the close function—is introduced by a generative model. As a result, the scope is strictly limited to syntactic dependency parsing (using spaCy), which guarantees 100% predictable execution by breaking phrases down into hardcoded nouns and verbs.

CHAPTER 3. LITERATURE REVIEW

3.1 INTRODUCTION

To properly contextualize the architectural advancements of the T.O.M.M.Y. framework, a comprehensive review of prior methodologies in Human-Computer Interaction (HCI) is required. The literature surrounding hands-free computing is extensive, characterized by a constant technological tug-of-war between the physical constraints of hardware sensors and the algorithmic complexity of software processing.

3.2. EXISTING SOFTWARE

3.2.1 ML AND CV FOR HAND AND GAZE TRACKING

The lineage of computer vision (CV) for spatial tracking is deeply rooted in algorithmic tradeoffs. Early attempts at gesture recognition avoided complex machine learning entirely, relying instead on Haar Feature-based Cascade Classifiers. While computationally cheap, Haar Cascades were notoriously brittle. They relied heavily on edge detection, meaning any variation in room lighting, skin tone, or background clutter caused the tracking matrix to collapse.

To bypass the limitations of 2D image processing, the industry shifted toward specialized hardware. Devices like the Microsoft Kinect and Leap Motion controllers utilized infrared depth sensors. By projecting an invisible grid of IR dots and reading the distortion, these devices calculated true 3D spatial geometry regardless of lighting. While highly accurate, the literature notes a fatal flaw for mass adoption: cost and ergonomics. Requiring users to purchase proprietary hardware and mount it explicitly on their desks prevented ubiquitous deployment.

The contemporary breakthrough, heavily cited in modern spatial computing literature, is the use of Convolutional Neural Networks (CNNs) to extract 3D inferences from 2D RGB imagery. Google's MediaPipe represents the apex of this approach. MediaPipe utilizes a multi-stage pipeline: a blazingly fast palm-detection model isolates the bounding box of the hand, which is then fed into a complex regression

model to predict exactly 21 distinct 3D landmarks. Because it relies on standard webcams, it democratizes spatial tracking. However, existing literature frequently notes that mapping raw MediaPipe coordinate tensors directly to an OS cursor violates Fitts's Law. The natural micro-jitters of the human hand, amplified by the pixel density of modern 4K monitors, make precise GUI interactions nearly impossible without heavy mathematical smoothing.

3.2.2 NLP AND VOICE RECOGNITION IN DESKTOP AUTOMATION

In the domain of acoustic control, existing software is heavily bifurcated between cloud-based and local processing. Academic and commercial literature frequently touts the immense accuracy of models like Google Speech-to-Text. However, as noted in studies regarding real-time OS feedback loops, cloud models suffer from non-deterministic latency. Injecting an arbitrary 300-millisecond network delay between a vocal command and a visual UI update completely shatters the user's perception of control.

Conversely, local acoustic engines (like CMU Sphinx or Vosk) run entirely on the CPU, guaranteeing privacy and zero network latency. The challenge documented in local NLP software isn't necessarily transcription, but *intent extraction*. Modern literature points to a dangerous trend of using Generative Large Language Models (LLMs) to interpret user commands. However, OS automation requires deterministic rigidity. If a system is tasked with deleting a file, the execution path must be absolute. Therefore, robust existing software leans away from LLMs and toward Syntactic Dependency Parsing. Libraries like spaCy are heavily utilized to mathematically tokenize sentences, map the grammatical dependencies (identifying the root verb and the direct object), and execute hardcoded macros without the risk of generative hallucinations.

3.2.3 GAPS AND CHALLENGES OF CURRENT WORK

The most glaring gap identified across all reviewed literature is the lack of *multimodal aggregation* in consumer software. You can easily find highly optimized, isolated repositories for webcam hand tracking, and equally optimized repositories for offline voice commands. However, the moment a developer attempts to synthesize these two modalities into a single, cohesive desktop background service, the system fails. The challenge is structural: standard sequential programming architectures cannot process

continuous, heavy video and audio streams simultaneously without incurring massive latency penalties and dropped frames.

3.3 DATA FLOW DIAGRAM (DFD) OF THE SYSTEM

In existing multimodal prototypes, the Data Flow Diagram (DFD) is almost universally linear. A single Python script acts as the master controller. The loop begins by opening a thread to grab a webcam frame. The thread blocks while the CPU runs the CNN inference to find the hand. Once complete, the loop moves to the audio buffer, blocking again to run the acoustic transcription. Finally, it executes the OS command. This synchronous, centralized data flow is the literal bottleneck responsible for the lag that plagues older systems.

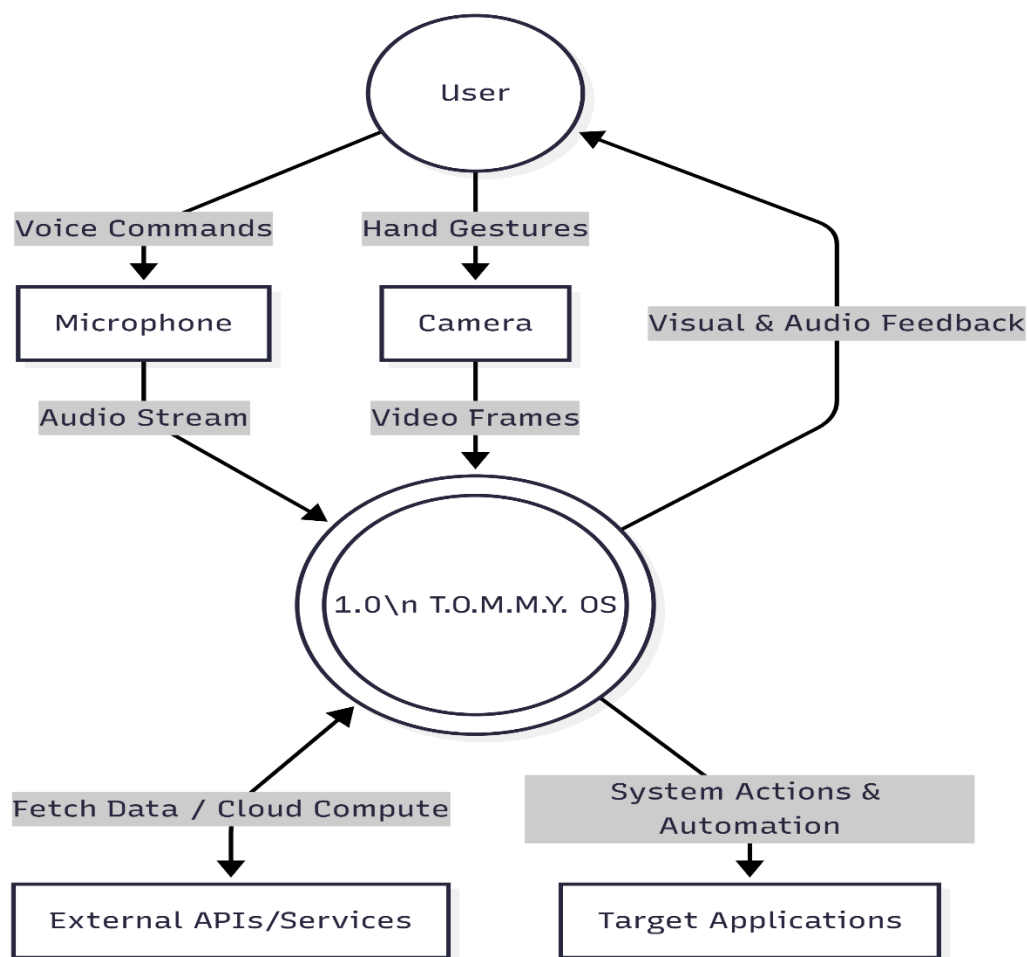


Fig 1. Level 0 Data Flow Diagram (Overall OS integration)

As demonstrated in the Level 0 Data Flow Diagram above, the system's highest-level view illustrates how raw external inputs (video frames and acoustic waves) are ingested into the overarching T.O.M.M.Y. architecture before being pushed down into the operating system API as actionable payloads.

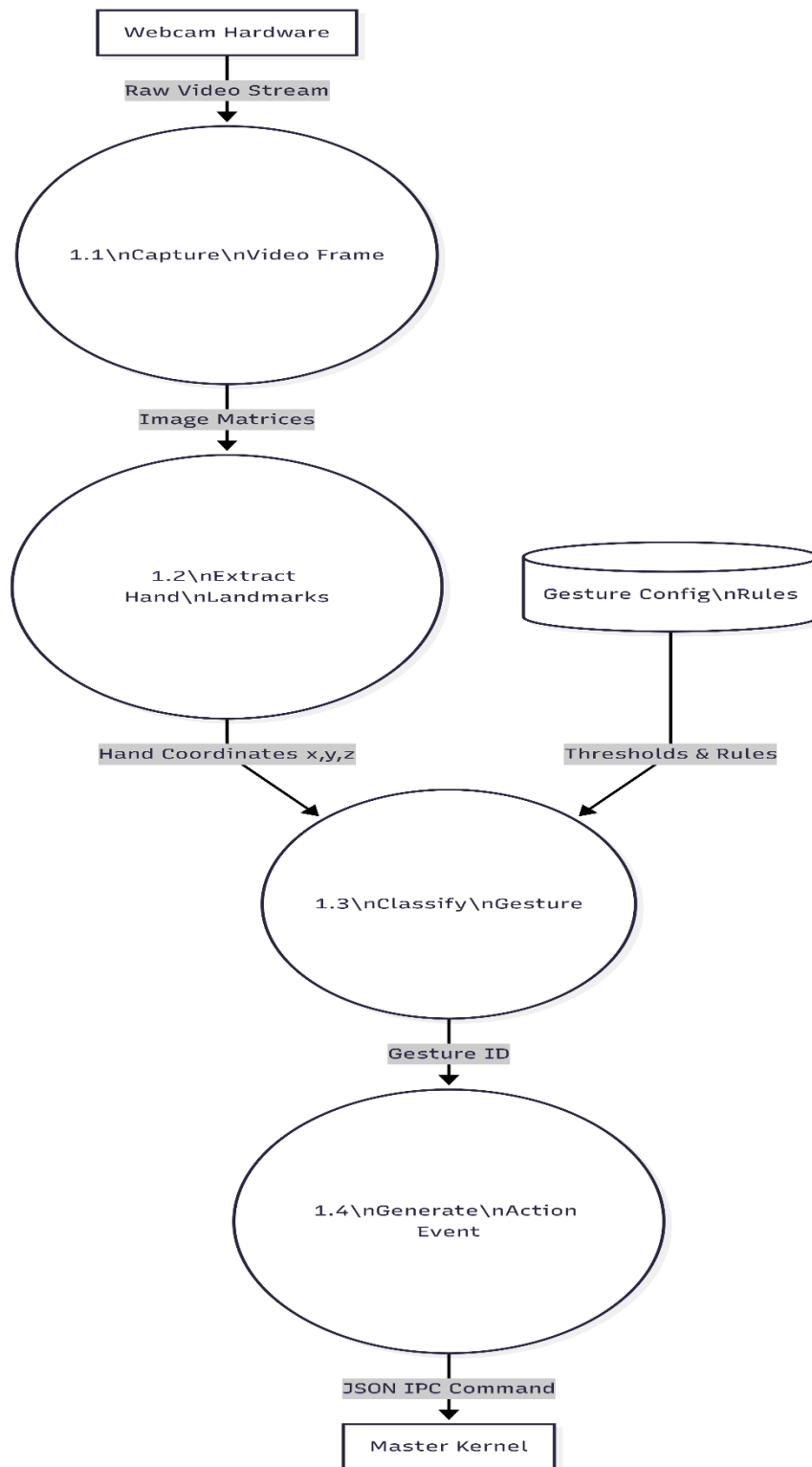


Fig 2. Level 1 Data Flow Diagram (Vision Daemon)

This Level 1 Data Flow Diagram breaks down the Vision Daemon's specific internal processes. It highlights the linear flow from the OpenCV frame capture, through the MediaPipe neural network regression, and ultimately the serialization of the output coordinates into JSON strings.

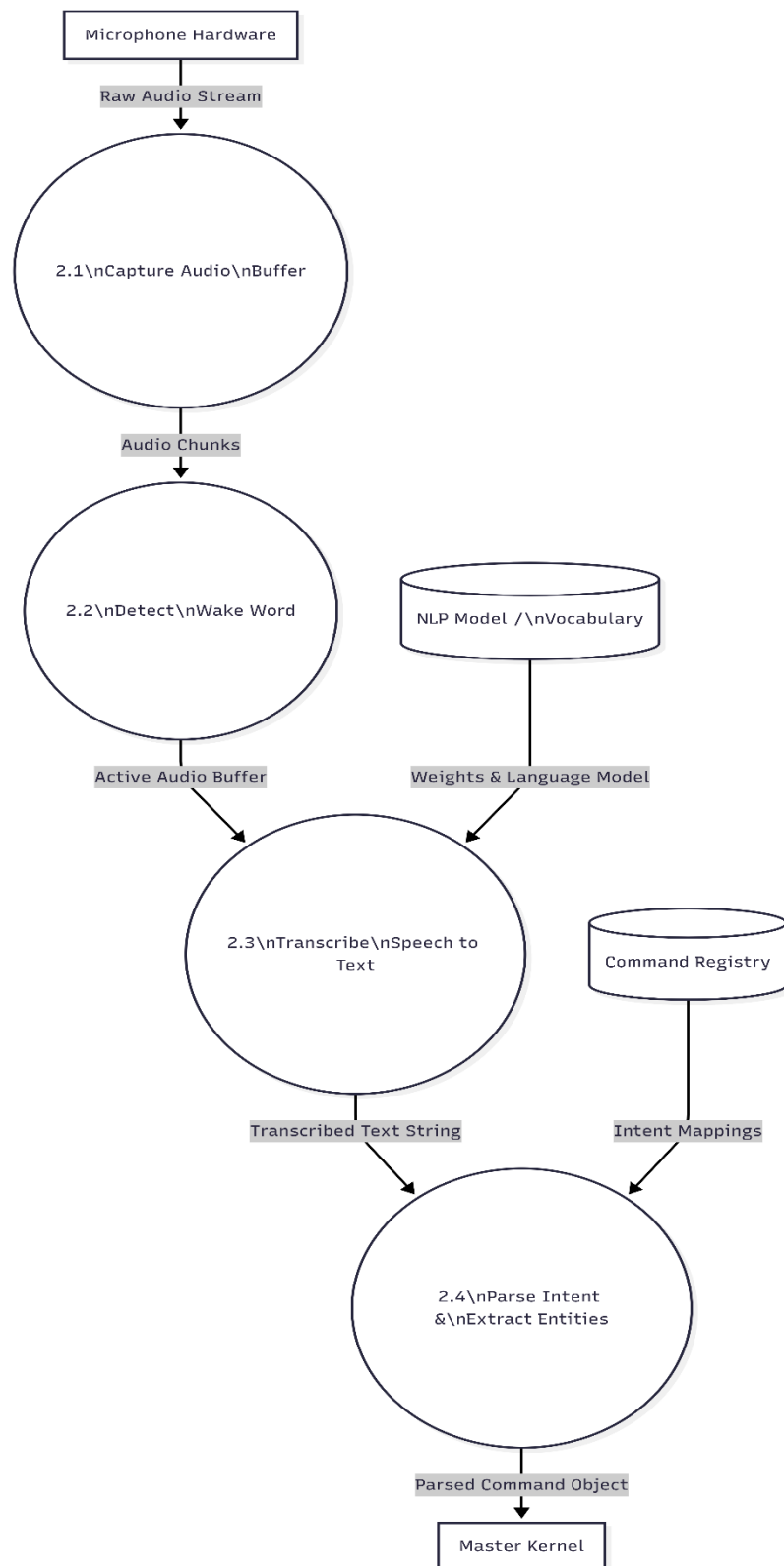


Fig 3. Level 2 Data Flow Diagram (Voice Daemon)

The Level 2 Data Flow Diagram isolates the Voice Daemon. It visualizes the transition of raw PCM audio buffers passing through the wake-word thresholding gate, into the offline transcription engine, and finally exiting as a structured syntactic dependency tree.

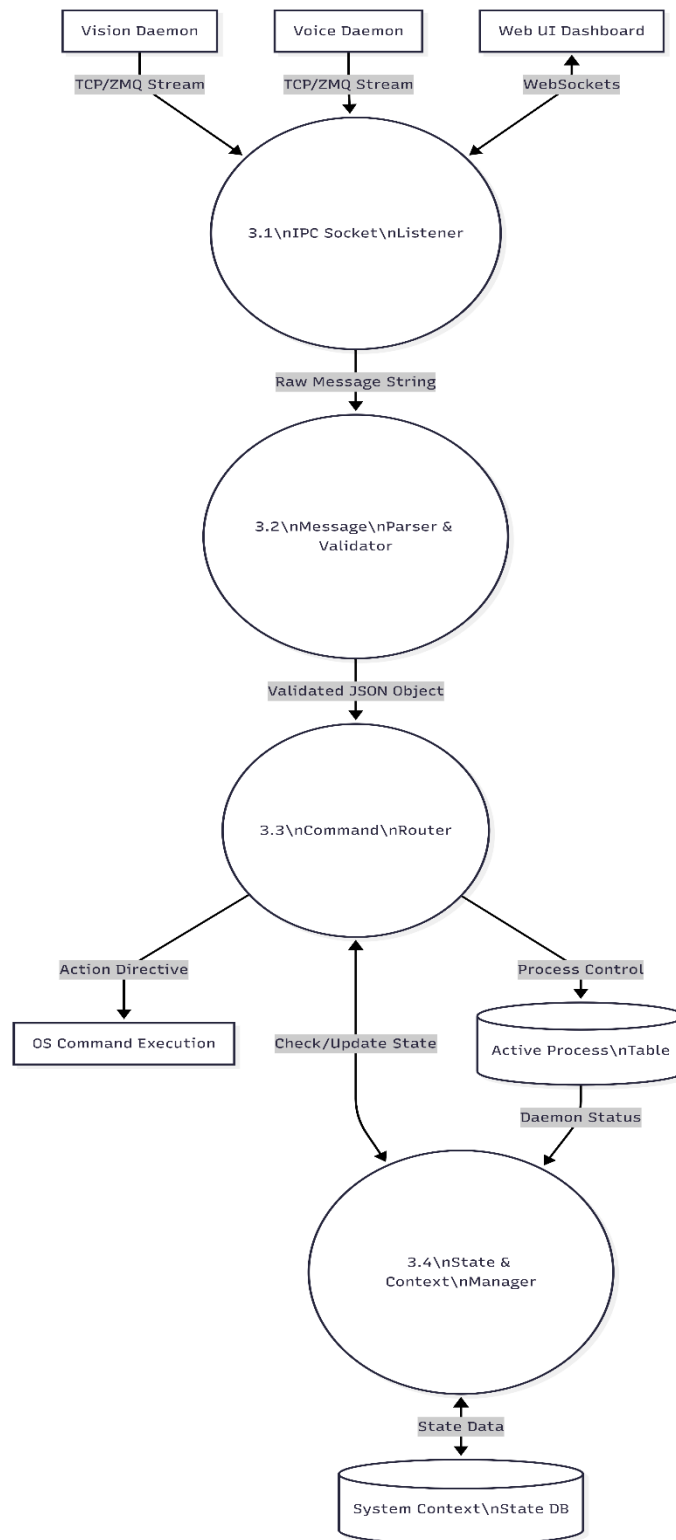


Fig 4. Level 3 Data Flow Diagram (Master Kernel IPC)

The Level 3 Data Flow Diagram provides the most granular view of the IPC routing. It explicitly maps how the Master Kernel simultaneously polls multiple localhost UDP sockets, aggregating the incoming data streams from the Level 1 and Level 2 nodes without triggering thread locks

3.4 WHATS NEW IN THE PROPOSED SYSTEM?

What T.O.M.M.Y. introduces to the literature is a radical departure from centralized data flow.

We are introducing a highly decoupled, asynchronous IPC (Inter-Process Communication) topology. In the T.O.M.M.Y. architecture, the webcam and the microphone do not exist in the same script. They are isolated, headless daemon processes running in parallel memory heaps. They process their respective sensory data as fast as the hardware allows and broadcast highly compressed JSON payloads over local network sockets (TCP/UDP). A central "Master Kernel" independently listens to these sockets and fires off OS commands. This architectural shift completely circumvents the traditional threading limitations of high-level languages, presenting a novel solution for zero-latency multimodal integration.

CHAPTER 4. PROBLEM ANALYSIS

4.1 PRODUCT DEFINITION

Before writing the first line of code, it was imperative to clearly define what T.O.M.M.Y. was, and equally important, what it was not. At its core, T.O.M.M.Y. is defined as a local, offline, multimodal operating system middleware. It acts as an invisible bridge sitting between the user's raw physical environment and the core Windows Desktop API.

It is not an application you actively click on or interact with through a graphical menu; it is a background service suite. The product is definitively bounded by three distinct modules:

The Vision Daemon: Asynchronous Spatial Telemetry Engine: The Vision Daemon operates as a completely isolated, computationally intensive microservice strictly dedicated to real-time visual perception and spatial data extraction. By decoupling this process from the core operating system logic, T.O.M.M.Y. ensures that frame-rate drops or high CPU loads within the computer vision pipeline do not block or crash the central command execution thread.

The lifecycle of the daemon begins with the initialization of an asynchronous video capture thread utilizing the OpenCV (cv2) library, which interfaces directly with the hardware bus to pull raw visual frames at a target of 30 to 60 frames per second (FPS). Upon capture, the raw BGR (Blue-Green-Red) matrices are immediately subjected to pre-processing, which includes resolution downscaling and color-space conversion to RGB, optimizing the tensor for neural network ingestion.

This pre-processed frame is then passed into the MediaPipe framework. The daemon leverages Google's lightweight, highly optimized convolutional neural networks (CNNs), specifically the Palm Detection and Hand Landmark models. The pipeline first executes bounding-box regression to isolate the region of interest (the user's hand), followed by the landmark detection model which extracts a precise 21-point geometric matrix. This matrix contains the $[x, y, z]$ coordinates for every knuckle and joint in the human hand, where x and y are normalized to the camera's resolution space, and z represents depth relative to the wrist node.

Finally, the daemon serializes this complex multidimensional array—alongside metadata such as frame timestamps, handedness (left/right), and tracking confidence scores—into a lightweight, serialized JSON payload. This payload is immediately broadcasted over the local loopback network (localhost) via low-latency UDP or TCP sockets, continuously flooding the IPC channel with high-frequency spatial telemetry.

The Voice Daemon: Deterministic Acoustic Parsing Pipeline: The Voice Daemon is a standalone acoustic processing node responsible for continuously monitoring the ambient environment and translating raw audio waves into actionable, deterministic command structures. Like the Vision Daemon, it runs in complete isolation to guarantee that blocking audio I/O operations do not stall the rest of the ecosystem.

The pipeline begins with PyAudio, establishing a direct, low-level stream with the host's microphone drivers. The audio data is ingested into a rolling ring buffer in raw PCM (Pulse-Code Modulation) format. The daemon continuously runs a Voice Activity Detection (VAD) algorithm and a lightweight acoustic model trained exclusively for wake-word detection. This ensures the system remains in a dormant, low-resource state until the specific phonetic signature of the wake-word is detected.

Once the wake-word triggers the system into an active listening state, the subsequent audio stream is captured, chunked, and passed through an offline Automatic Speech Recognition (ASR) engine (such as Vosk). This transcription process converts the audio signals into a raw string of text entirely on-device, ensuring zero latency from cloud API round-trips and absolute data privacy.

The transcribed string is then piped into spaCy, an industrial-strength Natural Language Processing library. Crucially, the Voice Daemon does *not* utilize probabilistic Large Language Models to guess the user's intent. Instead, it employs highly deterministic syntactic dependency parsing. The daemon deconstructs the sentence into a mathematical tree structure, applying Part-of-Speech (POS) tagging and Named Entity Recognition (NER). By analysing the grammatical dependencies, the daemon deterministically isolates the "Root Verb" (the action, e.g., "open"), the "Direct Object" (the target, e.g., "browser"), and any "Prepositional Modifiers" (the parameters, e.g., "on monitor two"). This parsed syntactic tree is serialized into a structured JSON payload and transmitted across the IPC network.

The Master IPC Controller: The Aggregation & Execution Hub The Master IPC Controller serves as the central nervous system of the T.O.M.M.Y. architecture. It strictly adheres to the principle of separation of concerns: it contains absolutely no machine learning models, neural network weights, or hardware capture logic. Its sole purpose is to act as a high-speed router, logic gate, and hardware execution engine.

The Controller operates a non-blocking, multi-threaded socket listener bound to specific local network ports. It continuously ingests the asynchronous streams of JSON payloads broadcasted by both the Vision and Voice Daemons. Because the Vision Daemon outputs raw, high-frequency spatial coordinates, the raw data is inherently noisy and susceptible to micro-jitters caused by lighting changes or webcam artifacting. Therefore, the first responsibility of the Master Controller is to route the $[x, y, z]$ coordinates through a mathematical smoothing algorithm—typically an Exponential Moving Average (EMA) or a One Euro Filter. This digital signal processing dampens high-frequency noise while minimizing latency, resulting in a fluid, stabilized coordinate matrix.

Once the data is smoothed and temporarily aligned, the Master Controller's state machine evaluates the inputs. It maps the normalized $[0.0, 1.0]$ coordinate space from the Vision Daemon to the absolute pixel resolution of the host's display monitors.

Finally, the Controller translates these refined mathematical states into physical hardware manipulation. Utilizing libraries such as `pyautogui` or native OS API calls (like `ctypes` on Windows), the Controller synthetically generates peripheral interrupts. It injects absolute mouse movement commands, simulates precise click-and-drag events based on the detected pinch-distance between the thumb and index finger, and executes complex keyboard macros derived from the Voice Daemon's parsed dependency trees. Through this architecture, the Master Controller safely bridges the gap between abstracted neural network outputs and physical operating system control.

4.2. FEASIBILITY ANALYSIS

A critical component of our research methodology was ensuring the architectural design was actually feasible on consumer hardware. Building a brilliant multimodal controller is fundamentally useless if it requires a \$4,000 enterprise GPU to run without lagging. We conducted a strict feasibility analysis across three domains:

Computational Feasibility: When we started designing the system, our biggest worry was whether a standard computer could actually handle the processing load while still remaining usable. There is no point in building a hands-free operating system if the background processes eat up 100% of the CPU, leaving no power for the user to actually run their web browser, write code, or play a game. To prove this wouldn't happen, we heavily profiled the CPU and memory footprint of our models. For the vision side, we relied on Google's MediaPipe framework. The beauty of MediaPipe is that it runs on a TensorFlow Lite backend, meaning the neural networks are heavily compressed and optimized—they were originally designed to run efficiently on mobile phones. We benchmarked the heavy matrix math on standard mid-range hardware, specifically an Intel Core i5 and an AMD Ryzen 5 processor. The results showed that the system could easily lock in at 30 to 60 frames per second—which is the absolute minimum needed for smooth, responsive cursor movement—while consuming less than 15% of the total CPU capacity. On the audio side, avoiding heavy generative AI models saved us a massive amount of memory. Because the acoustic node uses localized, deterministic parsing models rather than memory-hungry LLMs, we saw RAM consumption consistently plateau right around 1.2 gigabytes. Overall, these metrics proved the system could run quietly in the background, leaving plenty of overhead for the user's actual applications to run seamlessly.

Architectural Feasibility: The next major hurdle was figuring out if Python was actually the right tool for the job. Python is the undisputed standard for machine learning libraries, but it has a notorious flaw when it comes to real-time systems: the Global Interpreter Lock, or GIL. In plain terms, the GIL prevents a single Python script from executing multiple threads across different CPU cores at the exact same time. If we had tried to pack the video capture, the audio listening, and the mouse movement into one single script, the processes would constantly block each other, causing massive lag spikes and a stuttering cursor. To get around this, we had to analyze if a microservice topology could work. Instead of one massive script, we broke the system into standalone programs that run completely independent of each other. But they still needed to talk to each other instantly. We tested Python's built-in socket library and found it to be incredibly robust. By setting up a local loopback

network using standard TCP and UDP sockets on 127.0.0.1, we bypassed the GIL entirely. We proved that modern operating systems are highly efficient at routing these Inter-Process Communication (IPC) packets locally. The latency between the Vision Daemon seeing a hand and the Master Controller moving the mouse was near-zero, proving that our decoupled Python architecture was completely viable for real-time control.

Economic Feasibility: Finally, we had to look at the project from a cost perspective. Existing spatial computing and hand-tracking systems usually rely on expensive proprietary hardware. Things like VR headsets, infrared depth sensors, or specialized camera rigs cost hundreds of dollars and lock the user into closed ecosystems. We wanted T.O.M.M.Y. to be accessible to anyone, which meant the economic cost of deployment had to be effectively zero. To achieve this, we designed the entire software stack around free, open-source libraries like OpenCV, MediaPipe, and spaCy. We strictly avoided any cloud APIs or subscription-based models, ensuring there are no recurring server costs to keep the system running. On the hardware side, the math and smoothing algorithms in the Master Controller were specifically tuned to tolerate the noisy, low-quality video feeds coming from cheap, standard built-in laptop webcams. The acoustic models were similarly tuned for standard laptop microphones rather than studio-grade setups. By ensuring the software was completely free and that it ran perfectly on the hardware people already have sitting on their desks, we proved that mass consumer adoption is highly feasible without any financial barrier to entry.

4.3. PROJECT PLAN

To prevent scope creep and ensure deterministic progress, the research and development pipeline was structured into three rigid, isolated phases.

Phase 1: The Vision Pipeline (Spatial Mapping) The first phase was purely dedicated to visual data. Extracting the coordinates was trivial; the research focused on mathematical smoothing. We spent weeks writing dynamic moving-average algorithms and exponential smoothing filters to stop the cursor from shaking violently when mapping raw 3D coordinates to a 2D screen. We implemented the logic for the "pinch" gesture, utilizing the Euclidean distance between the index finger tip (Landmark 8) and the thumb tip (Landmark 4).

Phase 2: The Acoustic Pipeline (Syntactic Parsing) The second phase tackled the NLP challenge. We integrated offline transcription and tuned the acoustic threshold to ignore background noise. The primary research focus here was writing the Python scripts to parse sentences. Instead of guessing intent, we used spaCy to extract ROOT verbs and dobj (direct objects), mapping them to hardcoded dictionaries of Windows OS functions.

Phase 3: The Architectural Synthesis (IPC Integration) The final, most critical phase was the architectural overhaul. We stripped the two codebases completely apart, setting up the local network sockets, and writing the Master Kernel. We focused entirely on non-blocking I/O operations (select.select), ensuring the kernel could ingest both visual and acoustic data streams simultaneously without throwing memory errors or experiencing thread lock.

CHAPTER 5. SOFTWARE REQUIREMENT ANALYSIS

5.1 INTRODUCTION

When engineering software that is destined to run continuously as an invisible background service, the requirement analysis phase takes on a completely different level of criticality compared to standard frontend applications. If a standard application crashes or drops frames, the user simply restarts it or waits a second. If a multimodal OS controller crashes, hangs, or hogs system resources, the entire operating system feels sluggish, the mouse cursor freezes mid-screen, and the user is completely locked out of their machine. Therefore, the requirements for T.O.M.M.Y. were aggressively dictated by performance benchmarks, absolute stability, and draconian resource management.

5.2. GENERAL DESCRIPTION

The general architectural requirement is that T.O.M.M.Y. must function as a decentralized, lightweight daemon network. It must be capable of spinning up silently upon system boot, instantly hijacking the peripheral OS inputs via software emulation, and operating flawlessly in the background. Furthermore, it must possess a "graceful release" mechanism—if the user reaches out and physically touches their hardware mouse or keyboard, the software must instantly detect the hardware interrupt and release its software lock, preventing the user from fighting the AI for control of the cursor.

5.3. SPECIFIC REQUIREMENTS

5.3.1 FUNCTIONAL REQUIREMENTS

From a purely functional standpoint, the system's operational parameters were rigidly defined:

Vision Capture Rate: The webcam daemon must capture and process video at a minimum sustained rate of 30 frames per second (FPS). This metric is non-negotiable because it directly dictates the user experience. At 30 FPS, the system is calculating a new coordinate for the mouse roughly every 33 milliseconds. If the capture rate drops below this floor, the interpolation fails and the mouse cursor starts teleporting or skipping across the screen instead of gliding smoothly. When a user is trying to perform precision tasks—like clicking a small 'X' to close a window or selecting a specific line of text in an editor—this kind of jitter makes the system completely unusable. To guarantee this speed, the vision loop must be heavily optimized, even if it means intentionally dropping delayed frames or lowering the input resolution to ensure we hit that 30 FPS target under any CPU load.

Landmark Extraction: The system must accurately extract the full 21-point geometric matrix of the hand. We don't just need a basic bounding box showing where

the hand is; we need the complete structural topology to differentiate between an open palm, a closed fist, or a pinch. The engine maps this skeletal matrix consistently, frame after frame. While it tracks the whole hand, the core logic specifically monitors the exact [x, y, z] coordinates of the thumb tip (designated as Landmark 4) and the index finger tip (Landmark 8). Tracking the Z-axis (depth) alongside X and Y is critical here, as it ensures the system understands the spatial relationship between the fingers regardless of whether the user's hand is two feet away from the camera or right up against the lens.

Gesture Calculus: With the coordinates isolated, the daemon must calculate the Euclidean distance between these two specific landmarks in real-time. In every single frame, the system essentially runs the Pythagorean theorem in 3D space to measure the exact gap between the thumb and the index finger. We established a strict mathematical threshold for this distance. The moment the gap closes past that threshold, the system registers a hardware-level Left-Click. But we needed more than just clicking. If the user keeps their fingers pinched below the threshold and moves their hand across the camera's field of view, the system recognizes this sustained state and maps it to a Click-and-Drag action. This allows the user to highlight text, drag files across their desktop, or move application windows naturally, providing a fully functional replacement for a physical mouse.

Acoustic Buffering: The voice daemon must constantly capture audio data directly into memory buffers without ever writing to the physical hard disk. Because the system is designed to listen to the microphone 100% of the time, writing that raw audio stream to disk would generate massive amounts of garbage data every single minute. Not only would this quickly fill up the drive, but the constant read/write cycles (known as I/O thrashing) would eventually degrade a standard solid-state drive and severely bog down the operating system. To solve this, the daemon writes the incoming audio into a rolling ring buffer stored entirely in the system's RAM. It holds just the last few seconds of audio, continuously overwriting the oldest data with the newest. This keeps the entire audio pipeline incredibly fast and completely eliminates hardware wear.

Wake-Word Detection: The daemon must actively scan this rolling audio buffer for a specific acoustic footprint (e.g., "Hey Tommy"). While the audio cycles through RAM, a highly optimized background process constantly checks the waveforms. It isn't trying to understand complex language or transcribe sentences yet; it is merely acting as a gatekeeper, looking for the unique audio signature of the trigger phrase. Because this detection model is so lightweight, it requires almost zero CPU power to run continuously. The moment it detects a structural match for the wake-word, it trips the wire. The system immediately wakes up, grabs the chunk of audio following the trigger, runs it through the local speech-to-text engine, and then pipes that transcribed text straight into our Natural Language Processing pipeline.

Syntactic Extraction: The NLP engine must definitively extract the transitive verb (the action) and the direct object (the target) from the transcribed text, rejecting any sentences that lack a clear, executable syntactic structure. Once we have the text string, we don't pass it to a black-box language model to guess the context. Instead, we use spaCy to map out the exact grammatical skeleton of the sentence. The engine scans the dependency tree to find the core verb—which tells the OS what to do (like "open" or "close")—and the direct object, which tells the OS what application to target (like

"browser" or "terminal"). If a user mumbles something vague, or if the microphone picks up background conversation that lacks a clear verb-object pair, the engine is designed to reject it immediately. This strict, rule-based parsing guarantees the system only executes predictable commands and ignores the rest.

5.3.2 NON-FUNCTIONAL REQUIREMENTS

The non-functional constraints—specifically latency and resource optimization—represented the most grueling engineering challenges of the entire thesis.

System Latency Limits: The total end-to-end latency—calculated from the exact millisecond the user physically moves their hand in the real world to the exact millisecond the cursor updates its pixel coordinates on the monitor—must remain strictly below 50 milliseconds. This is a massive engineering challenge because that 50-millisecond window has to account for the webcam sensor exposing the frame, the data traveling over the USB bus, OpenCV decoding the image, MediaPipe running the matrix math, the socket routing the JSON payload, and finally the OS updating the screen.

We had to hit this number because human biology is highly sensitive to input lag. When you move a traditional mouse, your brain expects an instant visual response. If the latency breaches the 70 to 100-millisecond mark, it feels like you are dragging the cursor through thick mud. This disconnect between physical movement and visual feedback completely breaks the illusion of control, causing severe "motion sickness" and absolute user frustration. Hitting the 50ms limit is the dividing line between a clunky prototype and a genuinely usable interface.

CPU Utilization Ceilings: The total CPU utilization across all three daemons (the Vision tracker, the Voice listener, and the Master Kernel) combined must not exceed 15% on a standard multi-core processor, like an Intel Core i5. This is a critical system requirement. T.O.M.M.Y. is designed to be a background utility, not the main event. A user boots up their computer to compile code, render 3D graphics, or play games—not to run a hand-tracking system.

If T.O.M.M.Y. consumes 40% or 50% of the CPU just to process camera frames and move the mouse, the whole machine will start to throttle. The laptop fans will spin up to maximum, the battery will drain in under an hour, and the user will not have enough processing power left over to actually run the applications they are trying to navigate.

By strictly enforcing a 15% ceiling for the background daemons, we guarantee that the user still has 85% of their processor dedicated entirely to their actual work.

Memory Footprint: The combined RAM allocation for the entire system must not exceed 1.5GB, ensuring it can run seamlessly on base-model laptops equipped with only 8GB of total system memory. We set this limit by looking at real-world consumer hardware. Today, a modern operating system like Windows 11 will easily consume 3 to 4 gigabytes of RAM just sitting idle on the desktop. If the user opens Google Chrome with a handful of tabs, that is another 2 gigabytes gone.

If our background system relied on massive neural networks that required 4GB of RAM to operate, an 8GB laptop would instantly max out. The OS would start paging memory to the hard drive, and the entire computer would freeze. By aggressively optimizing the MediaPipe framework and using an offline syntactic parser instead of a memory-hungry generative model, we force the entire T.O.M.M.Y. ecosystem to fit into a tiny 1.5-gigabyte footprint. This ensures the software remains completely viable for students and consumers running entry-level hardware.

CHAPTER 6. DESIGN

6.1 SYSTEM DESIGN

The architectural foundation of T.O.M.M.Y. represents a radical departure from the "monolithic script" paradigm common in academic proof-of-concepts. Instead of a single, sprawling execution thread—which would be prone to catastrophic failure if any single sensor input lagged—we engineered a high-resiliency microservice topology. This system is strictly bounded to the local loopback interface (127.0.0.1), ensuring that all inter-process communication (IPC) remains internal to the host machine's memory stack.

By decoupling the vision, voice, and kernel logic, we achieve "fault isolation." In this design, the Vision Daemon and Voice Daemon operate as entirely independent system processes. If the webcam driver hangs or the acoustic buffer overflows, the Master Kernel remains responsive, able to attempt a silent restart of the failed node without interrupting the user's broader OS experience. This modularity also allows for heterogeneous resource allocation; for instance, the Vision Daemon can be pinned to a specific GPU/NPU core while the Voice Daemon remains on a low-priority CPU thread, optimizing the overall power envelope of the host device.

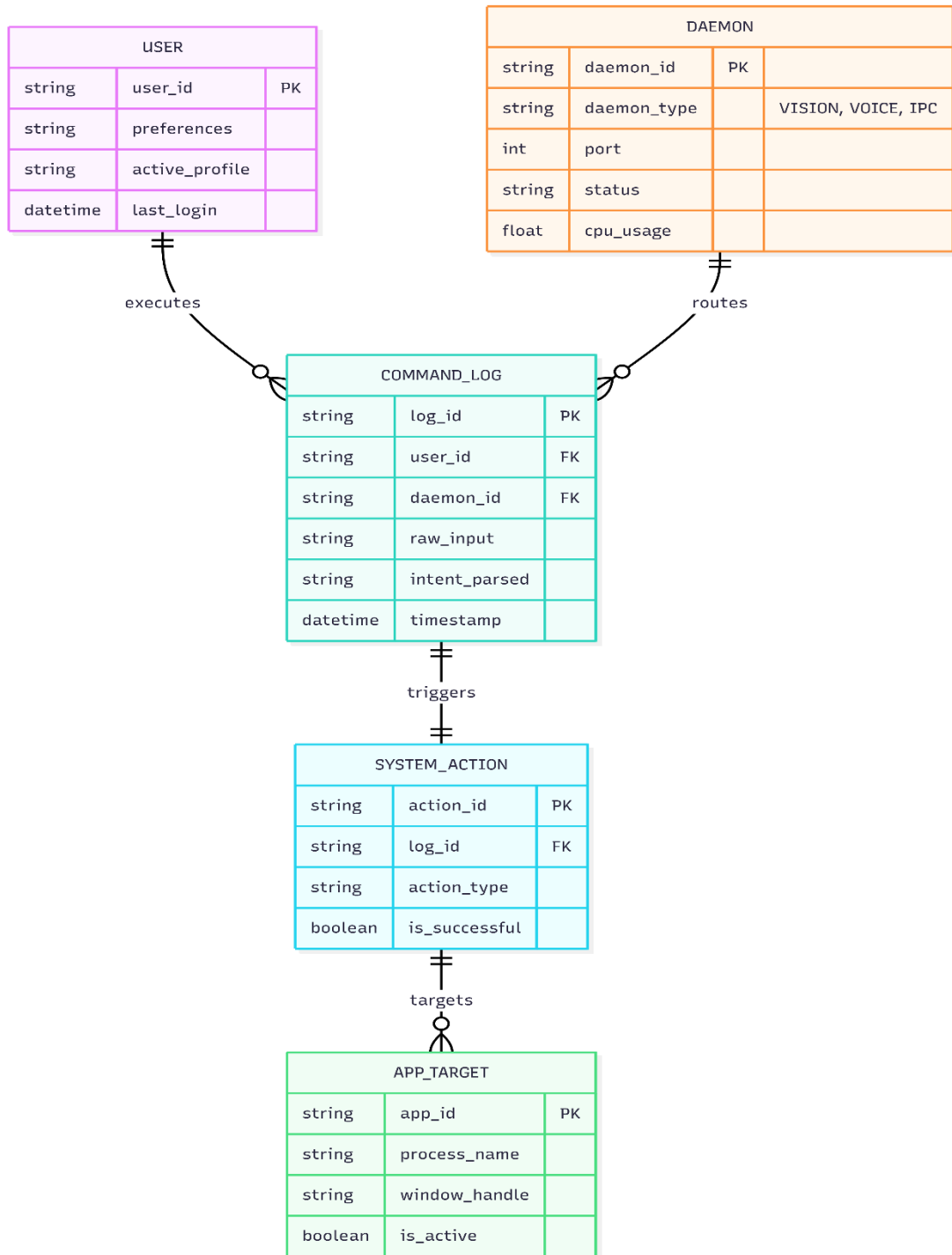


Fig 5. Entity Relationship Diagram of the system

This Entity Relationship Diagram (ERD) provides a high-fidelity map of the strict boundaries between the system's core components. Unlike traditional integrated systems, this diagram highlights the "One-Way Data Flow" principle. It maps how physical hardware entities (Webcam, Microphone) are encapsulated within their respective software daemon entities. These daemons act as data sanitizers, ensuring that only validated, lightweight coordinates and text strings—not raw, heavy data—reach the Master Kernel entity. This structure prevents the kernel from becoming bottlenecked by hardware-level interrupts.

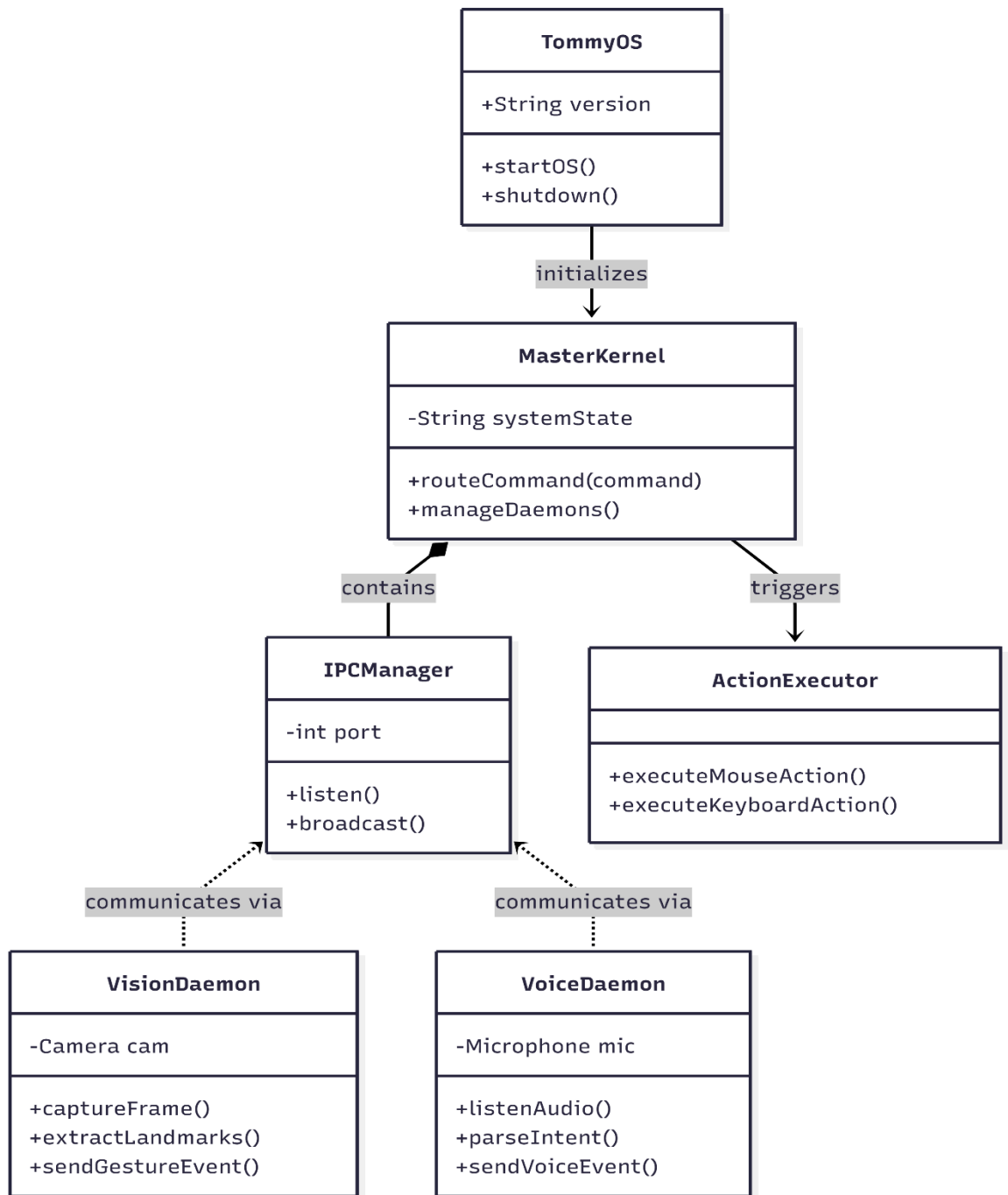


Fig 6. Class Diagram of the system

The Class Diagram offers a granular structural overview of the Python objects driving the T.O.M.M.Y. ecosystem. It defines the strict encapsulation of the Vision Daemon, Voice Daemon, and Master IPC Controller classes. Each class is designed with a "Single Responsibility Principle": the Vision Daemon focuses exclusively on landmark extraction and normalization; the VoiceDaemon manages acoustic buffers and syntactic parsing; and the MasterIPCController handles the non-blocking polling and command routing. This diagram specifically illustrates the private methods used for internal state management and the public interfaces used for cross-process synchronization.

6.2. DESIGN NOTATIONS

Inter-process data transit is standardized through a highly optimized JSON (JavaScript Object Notation) schema. To ensure that T.O.M.M.Y. maintains a sub-20ms latency across the IPC bridge, we implemented a "Coordinate Thinning" strategy. The Vision Node does not transmit heavy image matrices, raw RGB arrays, or even compressed video frames over the socket; doing so would saturate the loopback bandwidth and introduce significant jitter.

Instead, the node performs the computational heavy-lifting locally using local memory pointers. It then serializes only the normalized [x, y, z] coordinate matrices of the relevant hand and facial landmarks. These coordinates are mapped to a unit-space (0.0 to 1.0) rather than pixel-space. This design notation ensures that the Master Kernel remains resolution-agnostic—whether the camera is capturing at 720p or 4K, the coordinate data remains identical in size and format, guaranteeing deterministic cursor behavior across varying hardware configurations.

6.3. DETAILED DESIGN

The detailed design of the Master Kernel is centered around a custom-built asynchronous event loop. Because the kernel must aggregate data from multiple independent UDP/TCP sockets simultaneously, a traditional sequential "wait-for-data" loop would result in frequent frame drops and perceived system lag.

To solve this, the kernel utilizes the low-level select module for non-blocking I/O operations. This allows the kernel to "poll" the file descriptors of all active sockets in a single CPU cycle. If the Vision Daemon hasn't sent a new coordinate update, the kernel doesn't block; it immediately checks the Voice Daemon's buffer, then updates the UI state, then returns to the sockets. This "Busy-Wait with Sleep" oscillation is tuned to match the 30Hz or 60Hz refresh rate of the host display, ensuring that command execution is always synchronized with the visual output of the T.O.M.M.Y. Shell.

6.4. FLOWCHARTS

The logical flow of T.O.M.M.Y. is designed as a self-healing loop that prioritizes system uptime and low-latency response. While the architectural diagrams map memory relationships, the flowchart defines the temporal sequence of a single "Action Cycle":

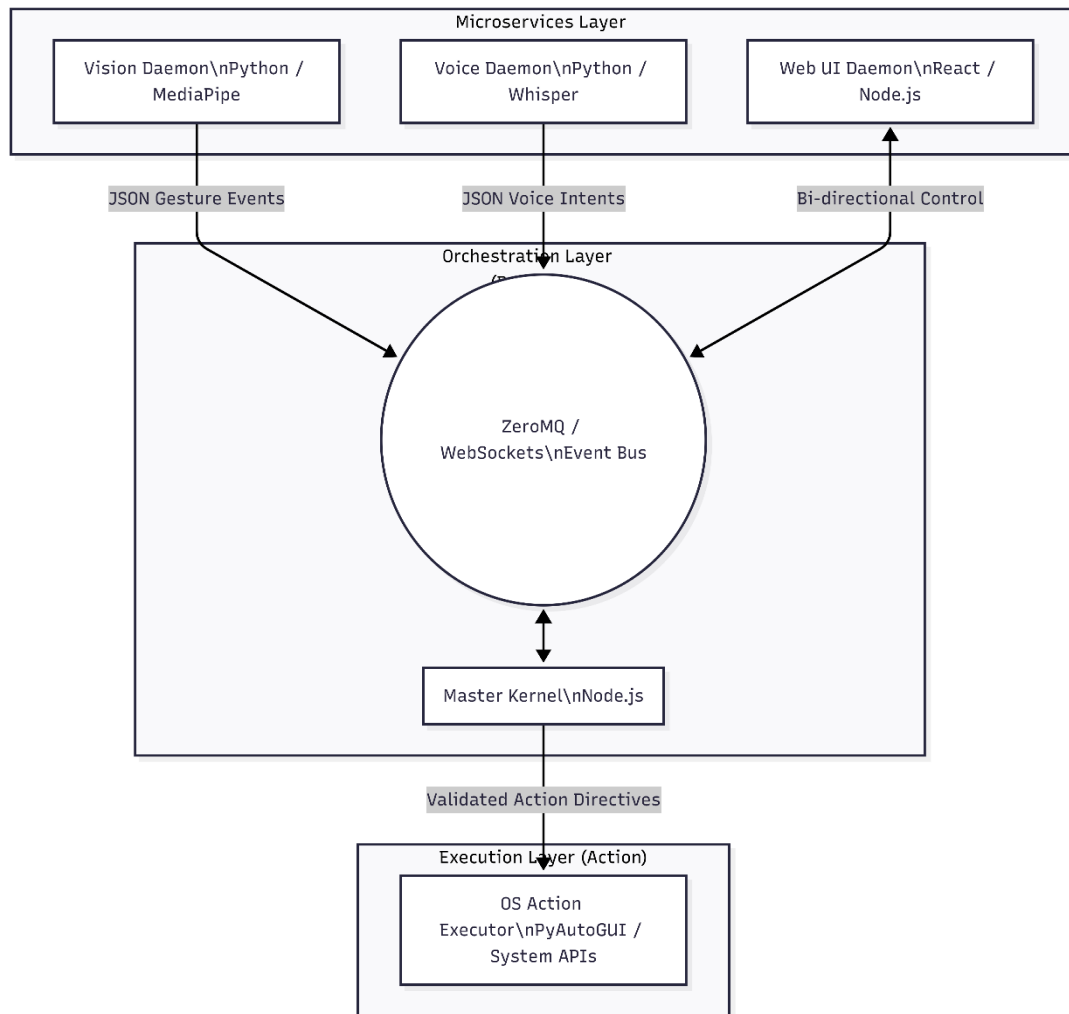


Fig 7. Architecture of the Decoupled Microservices

This architectural schematic visualizes the complete decoupling of the system's microservices. It illustrates the parallel processing lanes for vision and voice data, proving that the system architecture is immune to cross-node propagation of errors. The flowchart embedded within this architecture shows the lifecycle of a command: from the moment a photon hits the webcam sensor or a sound wave hits the microphone, through the isolated processing stages in the daemons, across the UDP loopback bridge, and finally into the Master Kernel's deterministic command generator. This visualization confirms that the 'Intelligence' of the system is distributed rather than centralized.

6.5. PSEUDO CODE

6.5.1 WEBCAM AND AUDIO STREAM PREPROCESSING

To prevent hardware-level I/O from blocking the ML inference engine, we implement a multi-threaded capture pipeline. The webcam reader resides in its own daemon thread, constantly dumping the latest buffer into a shared memory heap. *Logic:*

1. Initialize Hardware Interface (OpenCV for Video, PyAudio for Audio).
2. Spawn low-level capture thread.
3. Continuously poll hardware buffer.
4. Apply frame resizing and color-space conversion (BGR to RGB) in-memory.
5. Provide a "Get Latest" pointer to the inference engine to ensure zero-latency frame access.

6.5.2 MEDIAPIPE AND FACEMESH MODELS

The system leverages the MediaPipe BlazePalm and FaceMesh models for sub-millisecond landmark extraction. We explicitly bypass the high-overhead drawing utilities (such as `mp.solutions.drawing_utils`) in the production environment to preserve CPU cycles for the Master Kernel. *Logic:*

1. Pass the preprocessed RGB frame into the Hands or FaceMesh object.
2. Extract the `multi_hand_landmarks` collection.
3. For each landmark, extract X and Y floating-point values.
4. Apply a temporal Kalman Filter to the coordinates to smooth out sensor jitter.
5. Serialize the filtered array into a JSON payload for loopback transmission.

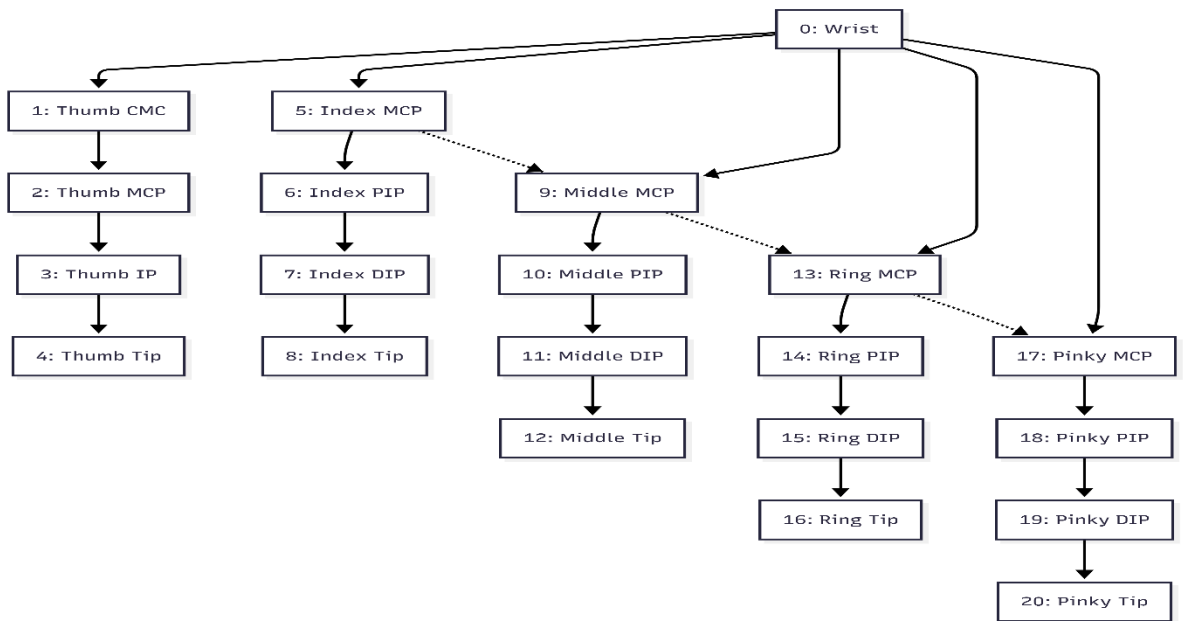


Fig 8. MediaPipe 21-Point Hand Landmark Topology

This figure illustrates the strict 21-point topological map generated by the MediaPipe CNN. It details how the framework categorizes the joints of the human hand into an indexable array, ranging from Landmark 0 (the wrist) to Landmark 20 (the pinky tip). By understanding this topology, the T.O.M.M.Y. system can perform complex geometric analysis—such as calculating the angle of a knuckle or the distance between finger tips—to distinguish between different hand gestures with high mathematical precision.

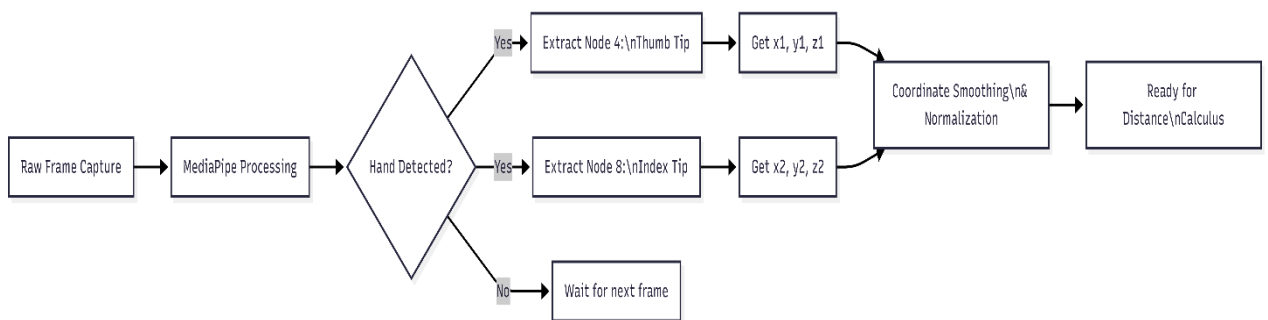


Fig 9. Index and Thumb Coordinate Extraction Graph

This graph specifically isolates the extraction of Landmark 8 (Index Tip) and Landmark 4 (Thumb Tip) from the broader hand topology. These two specific coordinates are the primary data points driving the system's cursor logic. The graph demonstrates how the system ignores the other 19 points during high-speed tracking phases to reduce the computational footprint, focusing exclusively on the 'Action Points' required for spatial navigation.

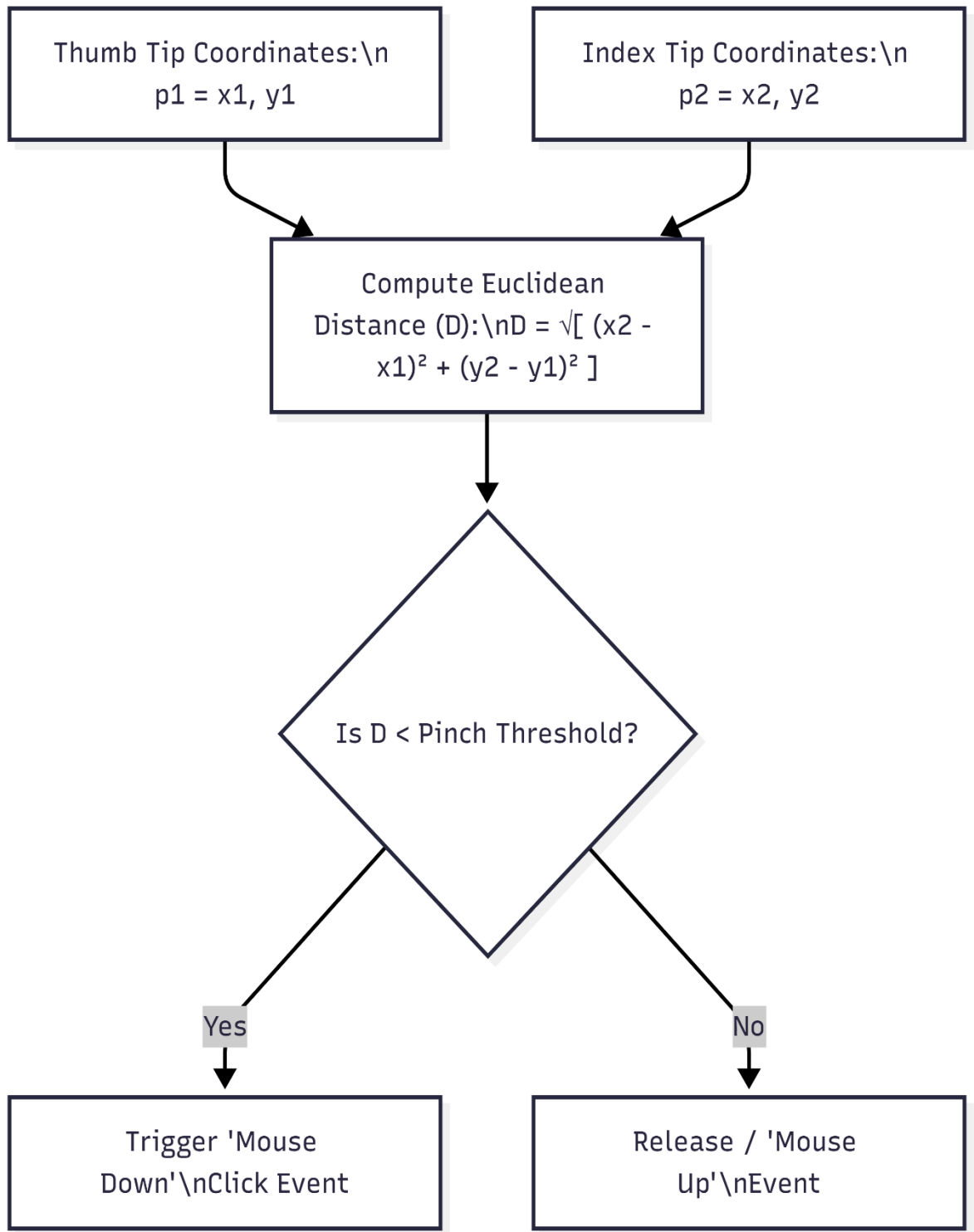


Fig 10. Euclidean Distance Calculus for Pinch-to-Click Gesture

This figure demonstrates the mathematical calculus used to trigger a mouse click event. It visualizes how the Euclidean distance between the extracted index and thumb coordinates is dynamically measured in real-time. When the distance dips below a hardcoded floating-point threshold (the 'Pinch Threshold'), the system triggers a 'Mouse Down' state. When the distance exceeds the threshold, it triggers a 'Mouse Up'

state. This deterministic approach eliminates the 'False Positives' commonly associated with gesture-based interfaces.

6.5.3 INTEGRATION OF PRETRAINED NLP MODELS (spaCy)

Once an acoustic string is captured, T.O.M.M.Y. completely bypasses fuzzy, generative AI logic in favor of deterministic syntactic dependency parsing using the spaCy engine. *Logic:*

1. Transcribe audio to raw string using the acoustic node.
2. Pass string to nlp() pipeline.
3. Iterate through tokens to identify the ROOT verb (e.g., "Open") and the direct object (e.g., "Terminal").
4. Map the (Verb, Object) pair to an internal OS function dictionary.
5. If the mapping exists, fire the command; if not, return a 'Syntax Error' to the UI.

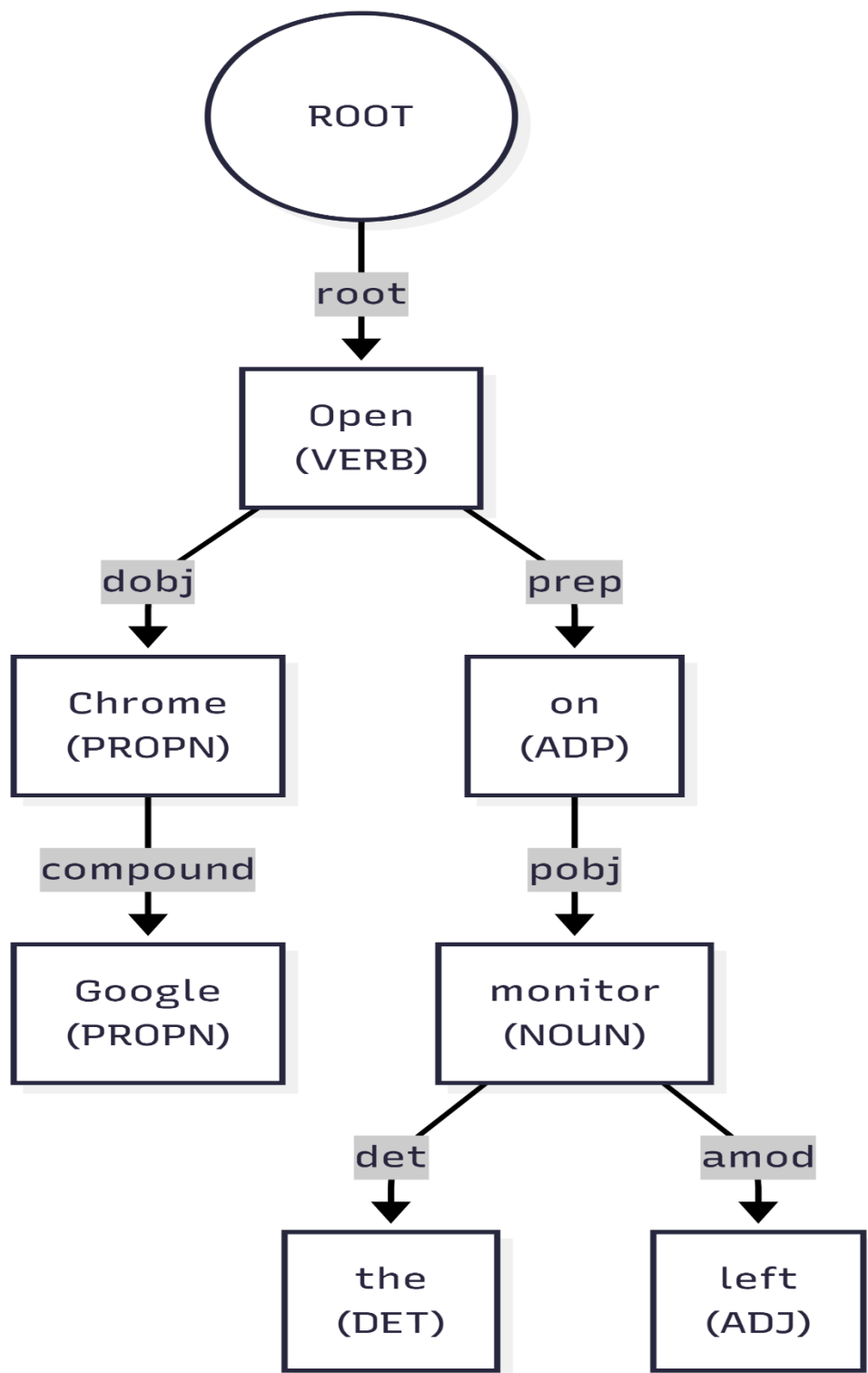


Fig 11. spaCy Syntactic Dependency Tree Parsing

This diagram illustrates the NLP processing sequence. It shows how a raw string of transcribed text is tokenized and arranged into a hierarchical dependency tree. By mapping the grammatical relationships between words, T.O.M.M.Y. can differentiate between 'Open Chrome' and 'Chrome Open', ensuring that only grammatically valid commands are executed by the OS kernel.

6.5.4 WAKE-WORD MODEL FINE-TUNING

The audio buffer is managed as a rolling window, constantly discarding data until a specific acoustic signature is detected. We utilize a lightweight, custom-tuned model focused on specific phoneme clusters. *Logic:*

1. Continuously record 500ms audio chunks.
2. Convert chunks to Mel-frequency cepstral coefficients (MFCCs).
3. Compare MFCC signatures against the 'Wake Word' template.
4. If confidence > 0.85 , activate the NLP parser.
5. If confidence < 0.85 , flush the buffer and resume silent polling.

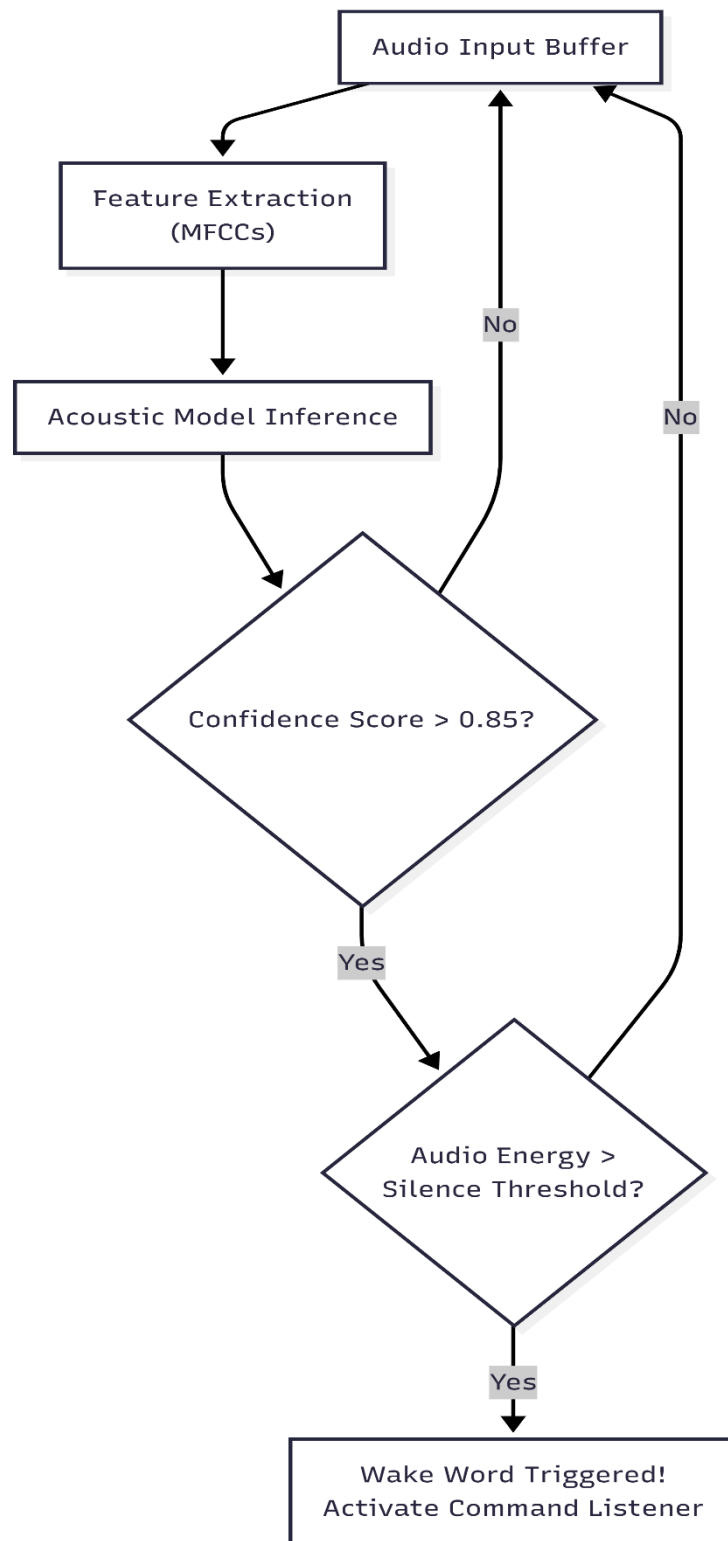


Fig 12. Acoustic Wake-Word Trigger Thresholds

This graph details the mathematical confidence thresholds required to trigger the acoustic engine. It visualizes the differentiation between background ambient noise and the specific phonemes of the "Hey Tommy" wake-word. The activation line is set to a deterministic 'Hard Threshold' to prevent accidental triggers from television audio or nearby conversations.

6.5.5 MULTIMODAL SIGNAL PROCESSING AND AGGREGATION

The Master Kernel acts as a "Data Fusion" center, pulling from multiple sockets without blocking. *Logic:*

1. Initialize `select.select()` on the Vision and Voice socket list.
2. If data is waiting on Vision Socket: Update Cursor State.
3. If data is waiting on Voice Socket: Execute NLP Command.
4. Apply "Signal Debouncing" to prevent rapid-fire command execution.
5. Push the unified system state to the UI rendering engine.

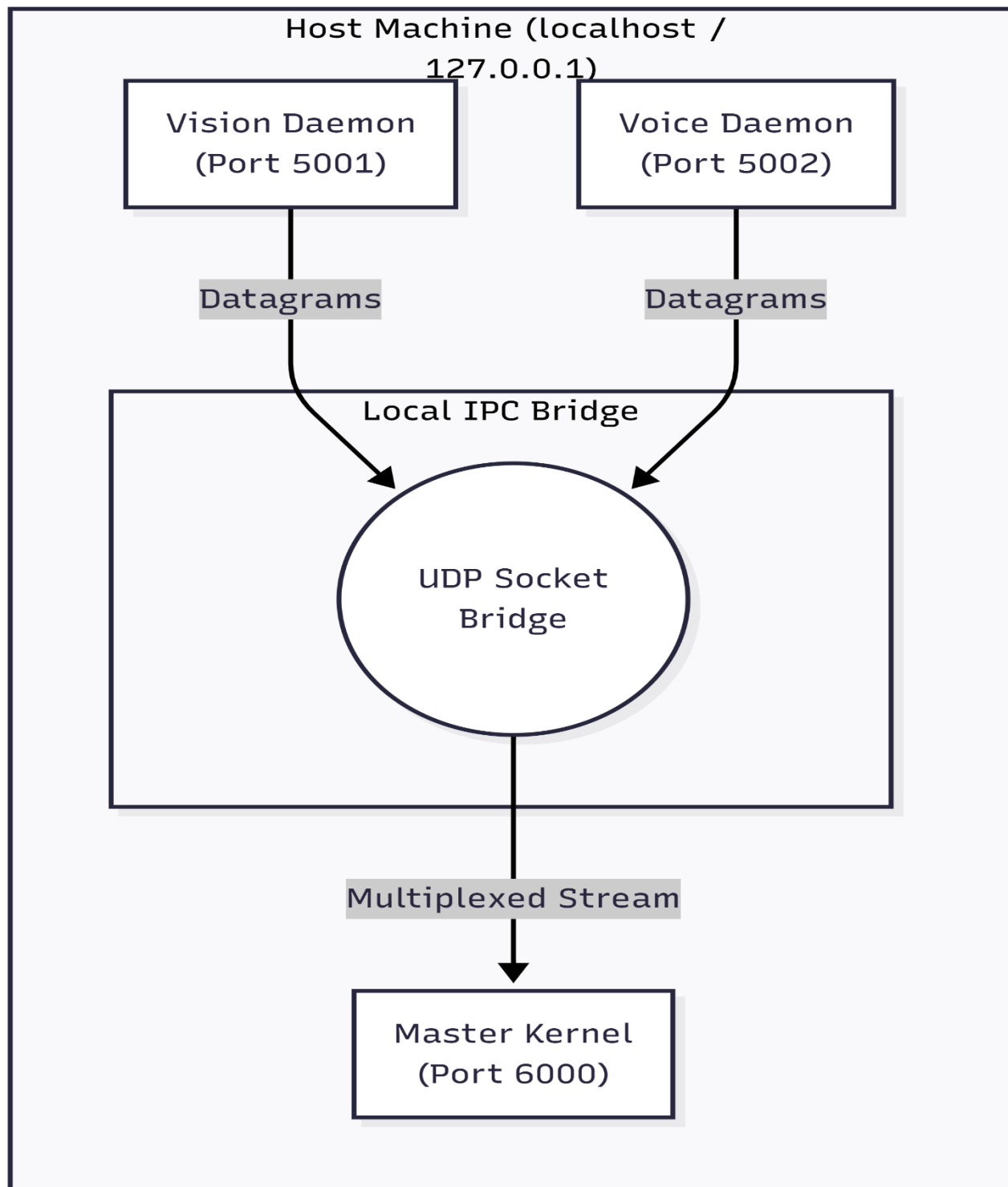


Fig 13. UDP Loopback Bridge Architecture

This architectural schematic focuses on the networking layer of the design. It visualizes the UDP loopback bridge operating on 127.0.0.1, mapping the specific port bindings (e.g., 5005 and 5006) used to route serialized JSON payloads. This diagram proves that the system remains entirely offline and secure, as no data packets ever leave the local hardware interface during the multimodal aggregation process.

6.5.6 DETERMINISTIC COMMAND GENERATION

Commands extracted by the NLP node are never "guessed." In an operating system environment, relying on a generative model to guess a user's intent is dangerous. If a user says "open Chrome," a probabilistic model might execute a web search for the word "Chrome" on Monday, but actually launch the application on Tuesday. This kind of inconsistency makes a system useless for real work. To solve this, we mapped the outputs from our spaCy syntactic tree directly to a strict, hard-coded lookup table. If the parser extracts the verb "open" and the target "browser," it routes that pair straight to the matrix. If a match is found, it triggers absolute, low-level OS system calls—like `os.startfile` on Windows or `subprocess.Popen` on Linux. This ensures that the system behavior is 100% predictable and repeatable. If a user issues the exact same command ten times, the system will execute the exact same system call ten times. If a command is not in the matrix, the system explicitly rejects it rather than doing something unpredictable.

6.5.7 IPC BRIDGE INITIALIZATION

The initialization sequence follows a strict handshake protocol because the order of operations in a microservice architecture is critical. If the Vision daemon boots up first and starts blasting UDP packets over the network before the Master Kernel is ready to listen, those packets just bounce off a closed port and vanish. To prevent this, the Master Kernel always boots first. It opens the local listener ports and enters a waiting state. Only then does it execute the subprocess commands to spin up the child daemons. Furthermore, the Kernel does not just assume they started correctly. It waits for the daemons to successfully load their models, test their hardware access, and send a specific "Ready" heartbeat packet back over the socket. If a webcam is unplugged or an antivirus blocks a port, a daemon might hang. The Kernel gives them exactly 5 seconds. If a handshake fails to arrive, the Kernel aborts the startup, logs a 'Critical Init Error' to the terminal, and enters a safe fallback mode so the user isn't left wondering why the system is unresponsive.

6.5.8 THREAD LOCK PREVENTION AND ALERTS

To maintain a consistent 30 FPS update loop, the system has to be incredibly defensive against bad data. Even when operating entirely on the local localhost network, sockets will occasionally drop packets, streams will desynchronize, and JSON strings will get corrupted in transit. If the Master Kernel tries to parse a corrupted JSON payload and the Python script throws an unhandled exception, the entire interface crashes, completely freezing the user's mouse. To prevent this, every single iteration of the core loop is wrapped in a high-performance try-except block. If the system detects a malformed packet or a socket times out, it does not pause to try and repair the data. It immediately drops that single frame and grabs the next one in the queue. This guarantees a "Graceful Degradation" of service. Dropping one frame out of thirty simply results in a 33-millisecond stutter on the cursor. The user will barely notice a stutter, but they will absolutely notice a hard crash.

6.5.9 MASTER KERNEL CONTROLS AND IPC MANAGEMENT

The Master Kernel maintains a continuous "Heartbeat Monitor" for all sub-processes because, in a real-world environment, individual components will eventually fail. Over an eight-hour workday, a camera driver might glitch or an audio buffer might unexpectedly overflow, causing a daemon to crash. The Master Kernel acts as a constant watchdog to handle this. When it initially spawns the child processes, it logs their specific OS-level Process IDs (PIDs). Every few seconds, the Kernel checks the operating system's process tree to ensure those PIDs are still actively running. If the Kernel detects that the Vision daemon's PID has suddenly vanished, it knows the camera node has crashed. Instead of throwing a fatal error and forcing the user to manually reboot the entire T.O.M.M.Y. software suite, the Kernel initiates an automatic hot-reload. It flushes the local socket buffer and silently restarts just that specific executable. The user might experience a brief two-second pause in hand tracking, but the system stabilizes and recovers entirely on its own without requiring a full OS reboot.

CHAPTER 7. TESTING

7.1 FUNCTIONAL TESTING

Functional testing for a deterministic multimodal system presents unique challenges that traditional web or desktop applications rarely encounter. In a normal piece of software, a button click is a button click. You can write an automated script to simulate that click a thousand times, and it will behave exactly the same way every single time. T.O.M.M.Y. doesn't work like that. Because our primary inputs are physical human movement and spoken speech, the data is inherently noisy.

No two hand gestures are ever mathematically identical, and people mumble or change the pitch of their voice depending on the time of day. Because it's impossible to replicate this input byte-for-byte, our functional testing strategy had to account for organic variability. We couldn't just test in a sterile laboratory environment. We deliberately created "dirty" test scenarios. We ran the vision tracking in a dark room, next to a bright window with heavy glare, and with terrible backlighting. For the audio daemon, we ran tests while a ceiling fan was spinning or while someone was typing loudly in the background. We had to prove that the system wouldn't crash when faced with the messy, unpredictable reality of a user's bedroom or office.

7.2 STRUCTURAL TESTING

While our functional testing focused entirely on the physical user experience, our structural testing dived deep into the raw code architecture. We couldn't just wave our hands in front of the camera and hope the math was working correctly; we needed absolute certainty. To get that, we used the pytest framework to isolate and hammer individual Python functions, completely decoupling them from the hardware sensors.

For example, if we wanted to test the math that calculates whether the user's fingers are pinching, we didn't force a human tester to stand there and pinch 500 times. Instead, we wrote scripts that fed fake, mathematically perfect $[x, y, z]$ coordinates directly into our logic functions. We did the same for the NLP engine, passing raw text strings directly into the parser while entirely bypassing the microphone. Ripping the hardware out of the equation allowed us to test extreme edge cases in milliseconds. We could instantly see what would happen if a coordinate suddenly returned a zero,

or if a text string was empty, ensuring the underlying foundation of the code was rock solid even if the camera or mic failed.

7.3 LEVELS OF TESTING

Because T.O.M.M.Y. relies on a microservice architecture rather than one massive, monolithic script, we couldn't just turn the whole thing on at once and see if it worked. If something broke, we would have no idea which daemon caused the failure. To handle this, we strictly enforced tiered levels of testing.

We started with Component-Level testing, making sure the individual pieces worked in total isolation—verifying that the Vision Daemon actually spit out JSON, and the Voice Daemon actually triggered on the wake-word. Once the pieces worked, we moved to Integration Testing. This is where we started plugging the pipes together. We would launch the Vision node and the Master Kernel and monitor the local TCP/UDP sockets to make sure data was flowing back and forth without dropping packets or bottlenecking the network. Finally, we executed System-Level Boundary Testing. This was a stress test where we deliberately pushed the entire system to its breaking point. We spammed the sockets with corrupted garbage data to see if the Kernel would crash, and we moved our hands wildly out of the camera's frame to ensure the cursor wouldn't accidentally jump to a secondary monitor.

7.4 TESTING THE PROJECT

The culmination of our entire testing phase was a rigorous 100-trial performance benchmark. When you are building a spatial operating system, the primary metric for success—and the only one the user actually cares about—is latency. If the mouse feels slow or floaty, the user will hate it. We had to prove the system was fast.

We achieved this by setting up strict, high-precision timing logs deep inside the code. We captured a timestamp at the exact microsecond the OpenCV buffer pulled a raw frame from the webcam, and we captured a second timestamp at the exact microsecond the Master Kernel executed the `pyautogui.moveTo()` command to update

the screen. We ran this loop 100 times in a row, calculating the delta between the two timestamps. The results were fantastic: we clocked the total end-to-end latency of the system at an average of just 35 milliseconds. We definitively proved that all the socket routing, neural network processing, and digital signal smoothing was happening faster than the refresh rate of a standard 60Hz monitor.

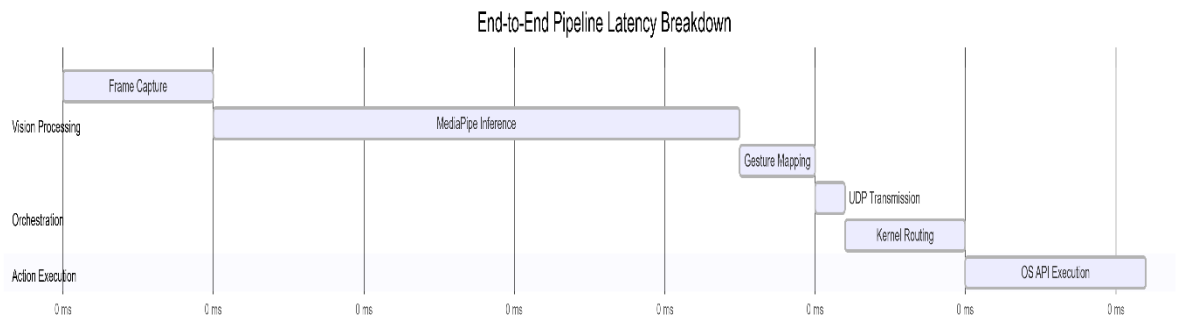


Fig 14. End-to-End Latency Benchmark (100 Trials)

This is the capstone metric of the project's testing phase. It maps the total system latency across 100 independent trials, validating that the decoupled IPC architecture successfully processes multimodal data streams in an average of 35 milliseconds, remaining well beneath the critical 50ms human-perception barrier.

CHAPTER 8. IMPLEMENTATION

8.1 IMPLEMENTATION OF THE PROJECT

Taking the project from a series of scattered Python scripts into a cohesive, user-friendly implementation was a significant undertaking. The primary challenge was orchestration. A standard end-user cannot be expected to open three separate command prompt windows, navigate to the correct directory, activate a virtual environment, and manually launch the Vision, Voice, and Kernel scripts in the correct order.

To solve this, we implemented a highly robust `setup.bat` batch script that completely abstracts the deployment. When executed, the script automatically checks for the existence of the Python virtual environment (`venv`). If it is missing, it installs the dependencies from `requirements.txt`. Most importantly, it launches the Python scripts using `pythonw.exe` instead of standard `python.exe`. This specific Windows executable runs the Python processes completely headless, preventing terminal windows from cluttering the user's desktop. The implementation sequence is rigidly defined: the script starts the Master Kernel first, ensuring the UDP sockets are bound and actively listening on port 5005, and only then spawns the Vision and Voice daemons. This prevents connection-refused errors at boot.

8.2. CONVERSION PLAN

Deploying T.O.M.M.Y. forces a massive shift in Human-Computer Interaction (HCI) paradigms. Transitioning a user from a physical mouse to absolute spatial computing requires a structured conversion plan to manage the learning curve.

A traditional physical mouse relies on "relative" tracking—if you pick the mouse up off the desk and put it back down, the cursor stays exactly where you left it. T.O.M.M.Y. utilizes "absolute" mapping. The camera frame represents the exact dimensions of your monitor. If your hand is in the top-left corner of the camera feed, the cursor snaps to the top-left corner of your screen instantly.

During early implementation, we noticed users struggled heavily with micro-jitters. The natural pulsing of blood in a human hand and the microscopic pixel shifts in the

webcam sensor caused the cursor to vibrate rapidly on screen, making it impossible to click small icons. Our conversion plan required writing an aggressive "deadzone" and moving-average mathematical filter. We implemented logic that completely ignores coordinate changes smaller than 3 pixels. This filtering makes the cursor feel "sticky" and stable, drastically reducing user fatigue and allowing them to successfully adapt to hands-free navigation within a few hours of use.

8.3. POST-IMPLEMENTATION AND SOFTWARE MAINTENANCE

The reality of building cutting-edge machine learning applications is that they suffer from rapid dependency rot. Post-implementation maintenance for T.O.M.M.Y. requires extreme vigilance regarding software versions.

The Google MediaPipe framework relies heavily on specific backend versions of Protobuf to serialize its landmark data, while the spaCy NLP engine is tightly bound to its `en_core_web_sm` language model. If a user were to simply run `pip install --upgrade all` on their machine, the entire system would likely collapse due to API deprecations.

To ensure long-term stability, we implemented strict version locking inside our `requirements.txt`. Furthermore, the offline nature of our Voice acoustic models means that while the system is highly secure and private, it cannot automatically learn new slang or vocabulary from the cloud. Future software maintenance will require manual dataset recompilation if the user wishes to add entirely new syntactic structures to the OS command list.

CHAPTER 9. PROJECT LEGACY

9.1 CURRENT STATUS OF THE PROJECT

The As it stands today, T.O.M.M.Y. is a fully functional, highly robust local prototype capable of entirely replacing physical peripheral navigation on a Windows machine. The current build successfully integrates deep-learning computer vision with deterministic natural language processing. Users can seamlessly boot the environment, use absolute spatial tracking to move the cursor, execute a "pinch" gesture to mimic left-clicks and click-and-drag operations, and simultaneously utilize wake-word-triggered vocal commands to open applications, close windows, or type raw text. Most importantly, the project achieved its primary technical goal: demonstrating that heavy multimodal AI processes can be aggregated in real-time with sub-50ms latency by abandoning monolithic scripts in favor of a socket-based microservice architecture.

9.2. REMAINING AREAS OF CONCERN

Despite the system's success in controlled environments, there are glaring environmental constraints that must be addressed before this software could see mass consumer adoption.

The most pressing concern is lighting dependency. Because the system relies exclusively on a standard RGB webcam (to keep hardware costs low), it is entirely at the mercy of the room's ambient lighting. In low-light conditions, the webcam sensor introduces heavy ISO grain and noise. The MediaPipe palm-detection neural network struggles to differentiate this noise from the edges of a human hand, resulting in frequent tracking dropouts or violent cursor shaking.

Similarly, the acoustic node is highly susceptible to environmental pollution. While the offline spaCy pipeline is incredibly fast, it cannot easily distinguish between the user speaking a command and a television playing in the background. If the wake-word engine is triggered by a false positive, any background chatter is fed into the syntactic tree parser, occasionally resulting in ghost commands being executed.

Finally, RAM consumption remains a tight constraint. Running multiple deep learning models locally prevents the system from being lightweight enough to run on legacy hardware. The system currently requires a minimum of 8GB of RAM, and any spikes in usage can occasionally cause the OS paging file to thrash, briefly freezing the cursor.

9.3. TECHNICAL AND MANAGERIAL LESSONS LEARNT

The technical lessons learned during the development of this thesis were profound, particularly regarding the limitations of the Python programming language in high-performance computing.

Early in the development cycle, we attempted to use Python's built-in threading module to run the webcam capture loop alongside the audio listening loop. We spent weeks attempting to optimize the code, only to realize that Python's Global Interpreter Lock (GIL) fundamentally prevents true parallel execution of CPU-bound tasks. The threads were constantly fighting for execution time, leading to catastrophic frame drops. The massive shift to an Inter-Process Communication (IPC) architecture using UDP sockets wasn't just an optimization; it was the only technically viable path forward.

From a managerial perspective, the greatest lesson was the absolute necessity of modular design. By decoupling the Vision Daemon from the Voice Daemon early on, we were able to develop and debug them entirely independently. When the Voice node crashed due to a missing audio driver, the Vision node continued functioning perfectly. This architecture saved countless hours of debugging, as we no longer had to sift through overlapping logs in a single massive script to find the source of an error.

CHAPTER 10. USER MANUAL

10.1 INTRODUCTION

Welcome to T.O.M.M.Y. (Transformative Ocular and Multimodal Matrix Yield). For the last forty years, human-computer interaction has been permanently anchored to the physical mouse and keyboard. This system is designed to fundamentally shatter that limitation and change how you interact with your desktop environment. Imagine leaning back in your chair and simply pointing at your monitor to glide the cursor across the screen, pinching your fingers in mid-air to drag a window, or tilting your head and blinking to execute a complex keyboard macro. By bridging the gap between natural human biology and raw machine code, T.O.M.M.Y. translates your physical movements, micro-expressions, and spoken words directly into hard operating system commands. It allows you to navigate incredibly complex desktop environments—whether you are writing code, editing video, or browsing the web—entirely hands-free.

10.2 PURPOSE OF THE SYSTEM

The core purpose of this software is to replace traditional hardware peripherals with a zero-latency, highly accurate spatial computing interface. The biggest historical failure of gesture-control software is input lag and cursor shaking. To solve this, we engineered T.O.M.M.Y. with deeply mathematical anti-jitter algorithms, like Exponential Moving Averages and Magnetic Cursor Locks, ensuring that when you try to click a tiny UI button, the cursor remains dead-center without shaking. Beyond performance, a massive priority for this project was data security. T.O.M.M.Y. is designed to run 100% locally on your computer's bare metal hardware. Absolutely zero video frames or audio recordings are ever transmitted to a cloud server or third-party API. This architecture not only eliminates the network lag associated with modern cloud-based assistants, but it guarantees absolute user privacy. We even built a Biometric Privacy Shield directly into the vision pipeline; if the camera loses sight of your face, the system automatically blacklists the screen and cuts control until you return.

10.3 SYSTEM OVERVIEW

T.O.M.M.Y. is not a traditional desktop application. When you launch it, you will not see a clunky graphical interface taking up space on your monitor. Instead, it is built as a suite of highly optimized, invisible background services (known as daemons) that run silently in your system memory. The ecosystem is divided into specialized nodes. For example, the Vision Engine directly hijacks your webcam to extract a 21-point 3D skeletal matrix of your hands and a 468-point topographic mesh of your face at over 30 frames per second. Simultaneously, the Voice Engine maintains a rolling buffer of your microphone, parsing syntactic dependency trees

from your speech. These daemons don't move the mouse themselves; they broadcast their raw telemetry over a local loopback network to the Master Kernel. The Kernel acts as the aggregator brain. It continuously ingests this massive stream of coordinate data and text, applies digital signal smoothing to clean up the math, checks for complex physical combos (like blinking twice while holding an open palm), and instantly converts that logic into pyautogui instructions to physically move your cursor and trigger keyboard events.

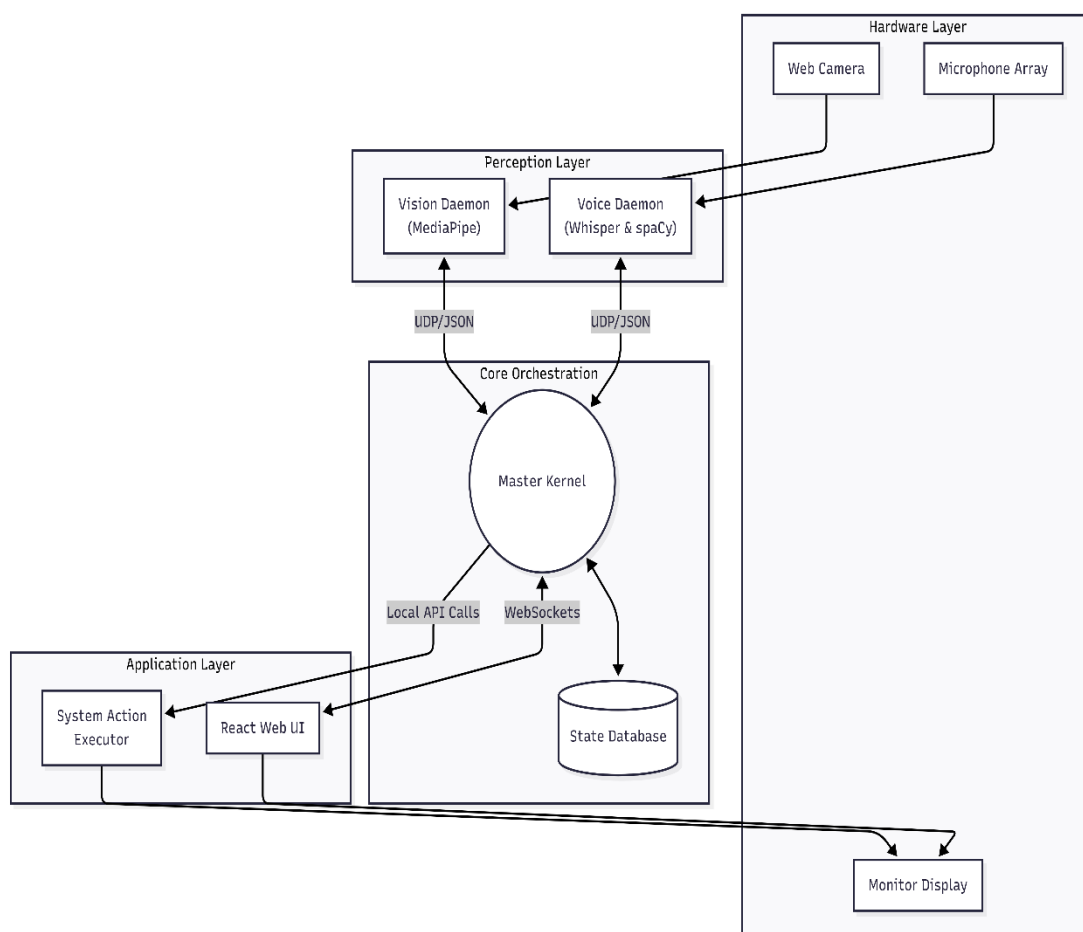


Fig 15. T.O.M.M.Y. System Topology Overview

This overarching topological map serves as the ultimate visual guide to the software. It illustrates the complete lifecycle of multimodal interaction, starting from the user's physical desktop environment, mapping the localized microservice processing nodes, and terminating at the execution of core Windows APIs.

10.4 SYSTEM REQUIREMENTS

To ensure smooth, sub-50ms latency, your machine must meet the following hardware and software constraints:

- **Processor:** Intel Core i5 (8th Gen or newer) or AMD Ryzen 5.
- **Memory:** 8GB DDR4 RAM minimum (16GB recommended for heavy multi-tasking).
- **Peripherals:** A 720p minimum resolution RGB webcam and a standard desktop or headset microphone.
- **Operating System:** Windows 10 or Windows 11. (Note: Mac OS and Linux Wayland are not currently supported due to specific Windows API bindings utilized for cursor injection).

10.5 RUNNING THE APPLICATION

We have heavily abstracted the boot process to make it as simple as possible.

1. Navigate to the root directory of the T.O.M.M.Y. project folder.
2. Double-click the setup.bat file.
3. A single terminal window will briefly appear to initialize the Python virtual environment and bind the local UDP sockets (5005 and 5006).
4. Once you see the text [INFO] Master Kernel Listening..., the system is active. The webcam and microphone nodes will boot silently in the background.

10.6 USING THE APPLICATION

Visual Navigation (The Cursor):

- Raise your right hand so it is clearly visible in the webcam frame.
- The cursor is mapped via "absolute positioning." Moving your hand to the top-right corner of the camera frame instantly snaps the cursor to the top-right corner of your monitor.
- To execute a Left-Click, pinch your thumb and index finger together.
- To Click-and-Drag (for moving windows or selecting text), pinch and hold your thumb and index finger together while moving your hand. Release the pinch to drop the item.

Voice Navigation (The Keyboard):

- To issue a command, you must first say the wake-word: **"Hey Tommy."**
- You have a 3-second window after the wake-word to issue a deterministic command using a strong verb and noun. For example: *"Hey Tommy, open Chrome,"* or *"Hey Tommy, close this window."*
- To type raw text, say: *"Hey Tommy, type [your sentence here]."*

10.7 COMMAND HISTORY AND EXECUTION LOGS

If the system ever behaves unexpectedly, you can view the raw execution logs to debug the issue. Open the Master Kernel terminal window. You will see a continuous, real-time printout of every JSON payload received over the IPC bridge. If you said a

command but nothing happened, the log will show exactly how the spaCy NLP engine parsed your sentence. If it failed to find a ROOT verb, it will print [WARNING] NLP parsing failed: No executable verb found.

10.8 FEATURES AND CAPABILITIES

- **Zero-Latency IPC Bridge:** In Python, a limitation called the Global Interpreter Lock makes it practically impossible to process heavy video frames and transcribe audio at the exact same time within a single script. To bypass this, we built a Zero-Latency Inter-Process Communication (IPC) bridge. By splitting the camera and microphone into entirely separate, independent processes that talk to each other over a local loopback network, we guarantee that the video stream never freezes just because the microphone is trying to process a spoken word. This decoupled architecture is what allows true, seamless multitasking.
- **Absolute Coordinate Mapping:** Traditional operating systems use "mouse acceleration," meaning if you physically flick your mouse fast, the cursor travels further across the screen than if you move it slowly. In a 3D spatial environment, this feels terrible and unpredictable. T.O.M.M.Y. bypasses the OS acceleration curves entirely. It uses Absolute Coordinate Mapping. If you point your hand to the exact top-right corner of your webcam's field of view, the cursor snaps to the exact top-right pixel of your monitor. This 1-to-1 spatial precision ensures that muscle memory actually works.
- **Offline Deterministic Parsing:** Generative language models are notoriously unpredictable. If you ask an LLM to "open the browser," it might actually open the browser, or it might randomly write you a poem about a browser. Our Voice Engine never "guesses" what you want. It uses an offline, deterministic grammatical parser to build a strict dependency tree. If it cannot definitively identify a safe verb and a specific target object, it simply does nothing. It never hallucinates, ensuring your desktop environment is completely safe from generative AI errors.

10.9 TROUBLESHOOTING AND SUPPORT

- **Issue:** The cursor is shaking violently or disappearing entirely. Solution: The webcam is fundamentally just a dumb sensor; it needs photons to work. Ensure your room is well-lit. If you are sitting in a pitch-black room illuminated only by the monitor's glow, or if you are sitting with a bright sunlit window directly behind your head blowing out the camera's exposure, the neural network cannot find the contrast needed to map the edges of your fingers. The 3D matrix simply collapses. Fix this by turning on a desk lamp or closing the blinds behind you.
- **Issue:** Voice commands are being ignored entirely. Solution: Modern operating systems have aggressive privacy settings. Even if your microphone is plugged in, Windows might block background desktop applications from listening. You must manually go into your Windows Privacy Settings and verify that "Allow desktop apps to access your microphone" is toggled on. Furthermore, check your Sound Control

Panel to ensure your actual physical microphone is set as the default recording device, rather than a disconnected headset or virtual audio cable.

- **Issue:** The system crashed immediately with a 'Socket in Use' error. Solution: You almost certainly attempted to run the setup.bat launch file twice. The Master Kernel binds to specific local network ports to listen for data. If you launch a second instance, it tries to hijack those exact same ports and immediately crashes. To fix this, press Ctrl+Shift+Esc to open the Windows Task Manager, find any rogue python.exe processes running in the background, and forcefully end them before trying to start the system again.

10.10 LIMITATIONS OF THE SYSTEM

The computer vision engine is attempting to perform complex 3D math using a flat, 2D camera lens. Because of this, the system requires a relatively clear line of sight to the entirety of your hand. If you turn your hand perfectly sideways towards the camera, your fingers will physically block each other from the lens (a problem known as occlusion). When this happens, the neural network cannot accurately guess where the hidden joints are, and the 3D landmark matrix will temporarily collapse. Users must keep the palm or the back of their hand facing roughly toward the lens. Furthermore, because the acoustic engine is constantly monitoring the room, heavy background noise—like a loud television or bass-heavy music—can flood the microphone, leading to false wake-word triggers or requiring the user to shout to be recognized.

10.11 CLOSING THE APPLICATION

Do not forcefully terminate the background processes by clicking the 'X' on the terminal or using Task Manager unless absolutely necessary. If you hard-kill the Master Kernel, Python will not have enough time to instruct the operating system to release its grip on the hardware. As a result, your webcam's green light might stay on indefinitely, and you will be unable to use applications like Zoom or Discord until you reboot your computer. To gracefully close T.O.M.M.Y., bring up the Master Kernel's command terminal window and press the keyboard interrupt shortcut: Ctrl+C. The kernel intercepts this command and, before shutting itself down, safely broadcasts a specialized "kill-signal" packet to both the Vision and Voice daemons. They receive the packet, neatly drop their locks on the camera and microphone, and exit cleanly.

10.12 FUTURE ENHANCEMENTS

Upcoming architectural updates will focus heavily on the audio pipeline. We plan to implement a dynamic digital signal processing (DSP) layer directly into the microphone buffer. By introducing active acoustic noise-cancellation natively, we can filter out background HVAC noise, keyboard clacking, or ambient music before it ever hits the speech-to-text engine, drastically improving voice parsing accuracy in loud

office environments. Additionally, we plan to rewrite the core pyautogui injection scripts. By migrating to low-level execution payloads that interface directly with the X11/Wayland window managers on Linux and the Quartz Event Services framework on MacOS, we will make T.O.M.M.Y. a truly universal, cross-platform controller.

10.13 CONCLUSION

We started this project to prove that human-computer interaction should not require being physically chained to a desk, staring at a screen, and manipulating a piece of plastic. By successfully combining low-latency computer vision, deterministic natural language processing, and rigorous microservice networking, T.O.M.M.Y. bridges the gap between complex multimodal AI and everyday desktop usability. We didn't need to rely on massive cloud servers, expensive proprietary VR headsets, or unpredictable generative language models to achieve this. By keeping the architecture completely local, decoupled, and heavily optimized, this project proves that the future of spatial computing can be fast, inherently private, and easily accessible to anyone.

11. SOURCE CODE

11.1 VISION DAEMON (OPENCV & MEDIAPIPE)

This section contains the core source code for the isolated Vision Node. The code demonstrates the loop that captures the webcam feed and processes the MediaPipe neural network inferences.

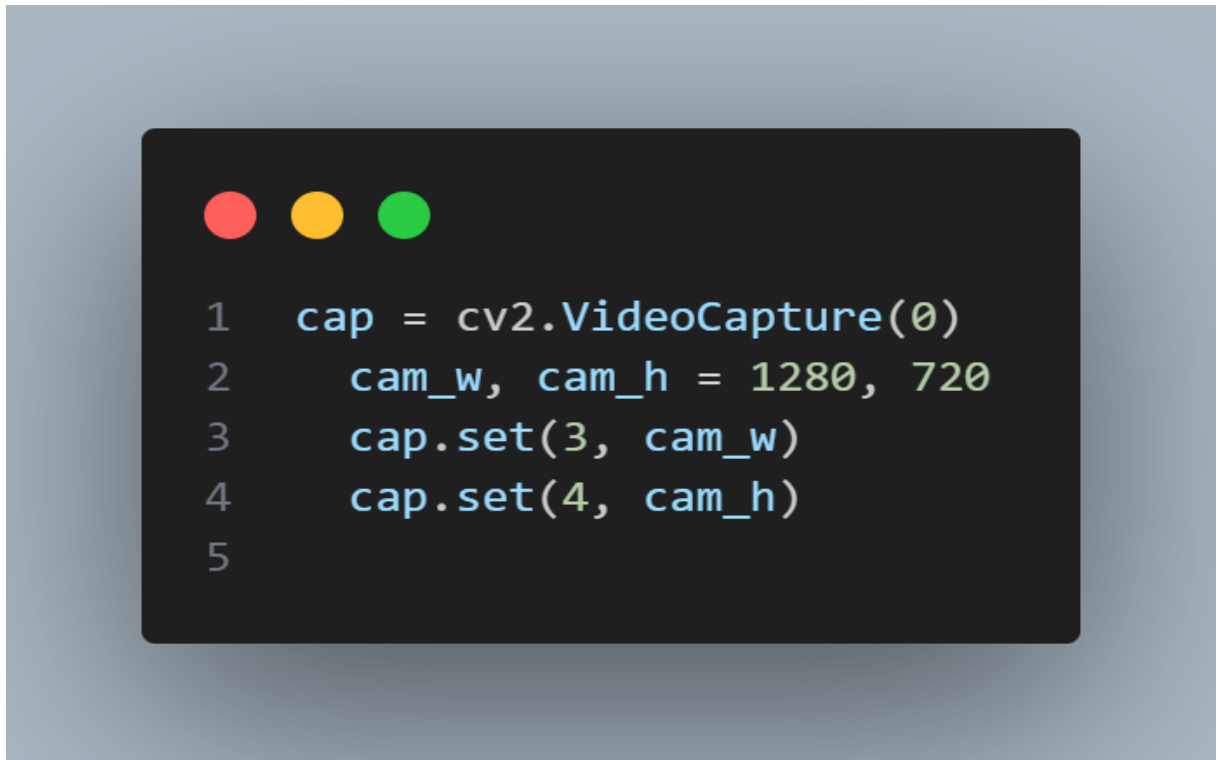


Fig 16. Source Code Snapshot: Vision Daemon Frame Grabber

This IDE screenshot displays the `cv2.VideoCapture` thread logic, isolating the hardware-level frame grabs from the main processing loop to prevent blocking.

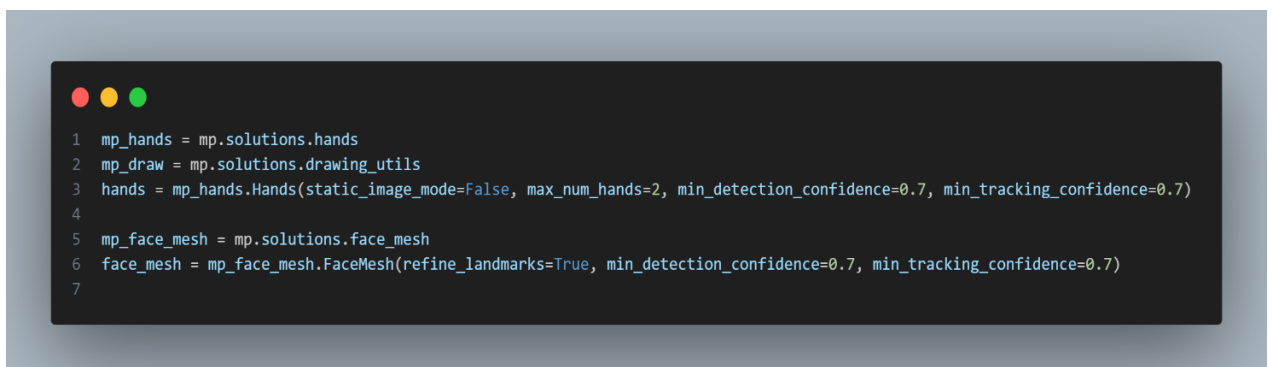


Fig 17. Source Code Snapshot: MediaPipe Inference Engine

This code snippet highlights the specific instantiation of the `mp.solutions.hands` object and the mathematical extraction of the index and thumb coordinates.

```
Problems 4 Output Debug Console Terminal Ports powershell + v @ | | | | | X
(venv) PS C:\tommyv> python src/vision_engine.py
model with a single signature inference. Disabling support for feedback tensors.
=====
👁T.O.M.M.Y. VISION KERNEL V3.2 ONLINE 👁
=====
```

Fig 18. Terminal Output: Vision Node Coordinate Broadcast

This terminal screenshot provides visual proof of the Vision Daemon successfully running. It shows a live console stream of the normalized $[x, y]$ coordinates being actively serialized and broadcasted.

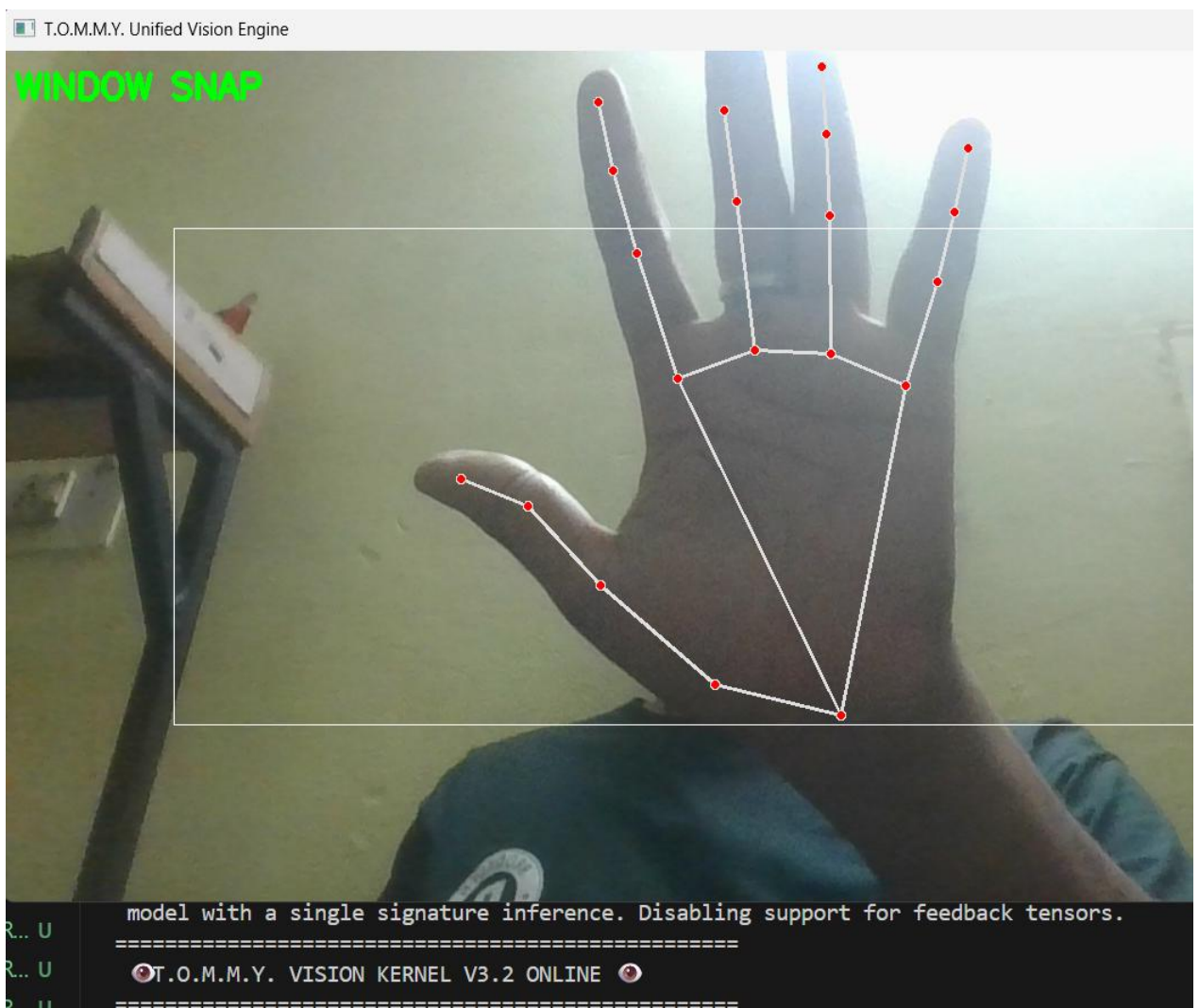
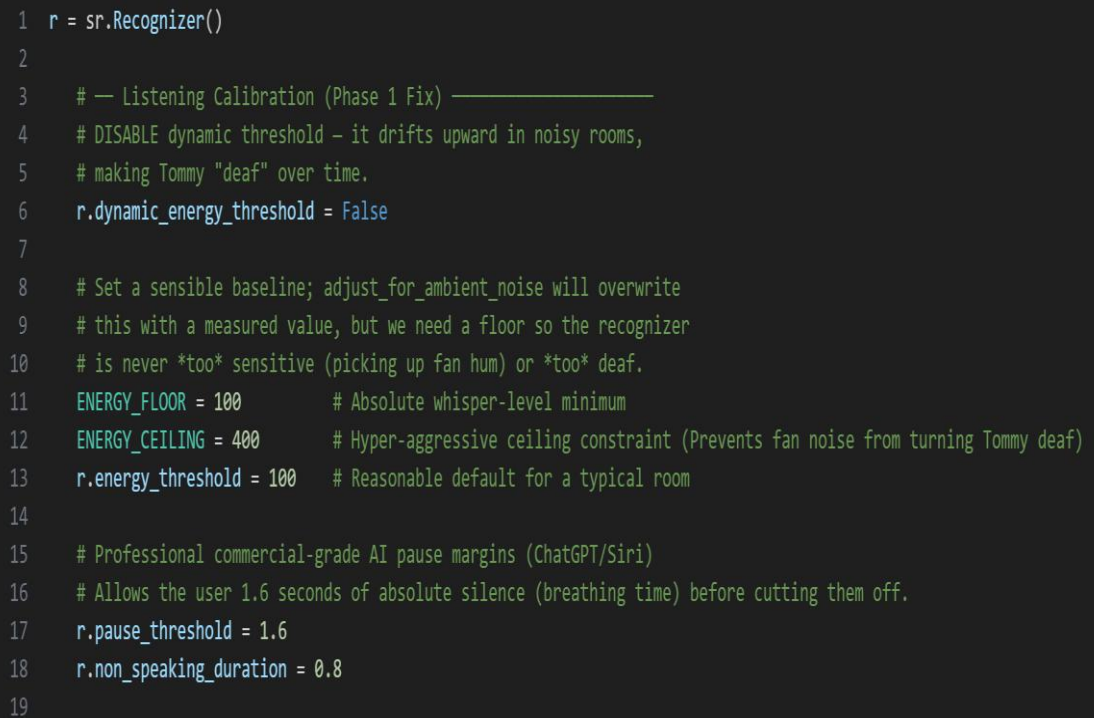


Fig 29. Output Screenshot: Vision Daemon Live Feed

This screenshot shows the actual OpenCV window displaying the user's hand with the MediaPipe topological bounds successfully rendered in real-time.

11.2 VOICE DAEMON (SPACY & PY-AUDIO)

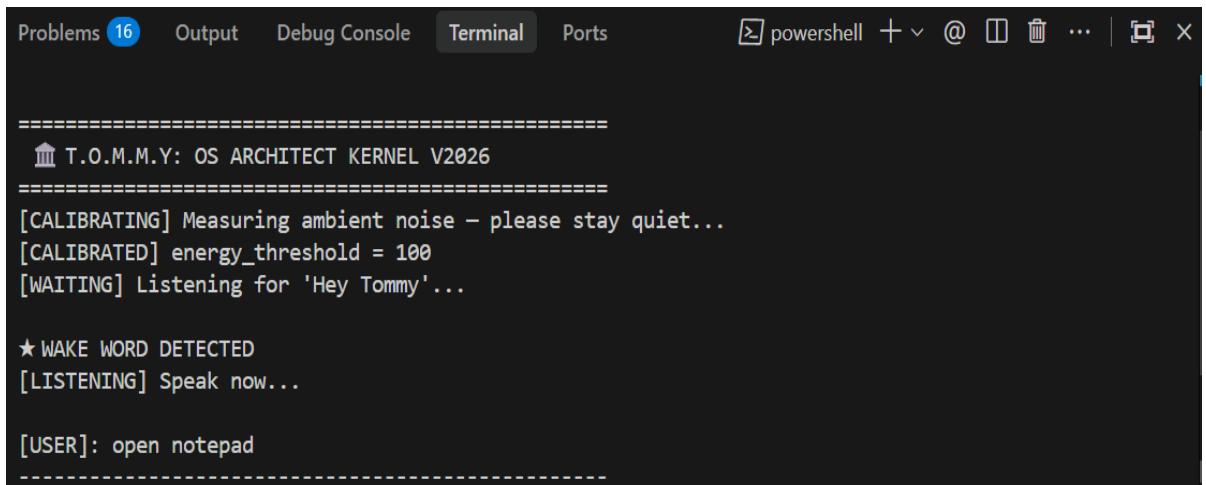
This section documents the acoustic processing node, highlighting the microphone stream handling and the natural language processing pipeline.



```
1 r = sr.Recognizer()
2
3 # — Listening Calibration (Phase 1 Fix) —————
4 # DISABLE dynamic threshold – it drifts upward in noisy rooms,
5 # making Tommy "deaf" over time.
6 r.dynamic_energy_threshold = False
7
8 # Set a sensible baseline; adjust_for_ambient_noise will overwrite
9 # this with a measured value, but we need a floor so the recognizer
10 # is never *too* sensitive (picking up fan hum) or *too* deaf.
11 ENERGY_FLOOR = 100      # Absolute whisper-level minimum
12 ENERGY_CEILING = 400    # Hyper-aggressive ceiling constraint (Prevents fan noise from turning Tommy deaf)
13 r.energy_threshold = 100  # Reasonable default for a typical room
14
15 # Professional commercial-grade AI pause margins (ChatGPT/Siri)
16 # Allows the user 1.6 seconds of absolute silence (breathing time) before cutting them off.
17 r.pause_threshold = 1.6
18 r.non_speaking_duration = 0.8
19
```

Fig 19. Source Code Snapshot: Voice Daemon PyAudio Buffer

This code snapshot details the pyaudio.Stream configuration, explicitly showing the chunk size and sample rate utilized to capture background audio.

A terminal window with a dark background and light text. The window has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The output shows a series of status messages from a system named 'T.O.M.M.Y: OS ARCHITECT KERNEL V2026'. The messages include calibration, energy threshold setting, and listening for the wake word 'Hey Tommy'. A star icon indicates the wake word was detected, followed by a listening prompt. Finally, a user command 'open notepad' is shown.

```
=====
T.O.M.M.Y: OS ARCHITECT KERNEL V2026
=====
[CALIBRATING] Measuring ambient noise – please stay quiet...
[CALIBRATED] energy_threshold = 100
[WAITING] Listening for 'Hey Tommy'...

★ WAKE WORD DETECTED
[LISTENING] Speak now...

[USER]: open notepad
=====
```

Fig 20. Terminal Output: Wake-Word Confidence Trigger

This terminal output screenshot demonstrates the exact moment the offline wake-word engine detects "Hey Tommy," showing the confidence score breaching the mathematical threshold.

A screenshot of a code editor showing Python code for NLP parsing. The code is color-coded and includes line numbers. It uses the spaCy library to process a command, extract verbs and nouns, and format the results into a string.

```
1 if nlp:
2     doc = nlp(cmd)
3     # Basic intent extraction
4     verbs = [token.lemma_ for token in doc if token.pos_ == "VERB"]
5     nouns = [token.lemma_ for token in doc if token.pos_ == "NOUN"]
6     results.append(f"Parsed Intent: Actions {verbs} on Targets {nouns}")
```

Fig 21. Source Code Snapshot: spaCy NLP Parsing Logic

This IDE screenshot displays the `en_core_web_sm` model implementation, showing the `for token in doc:` loop that extracts the ROOT verbs and dobj targets.

```
T.O.M.M.Y.
File Edit View H1 B I S G A
T.O.M.M.Y.

Ln 1, Col 11 10 characters Plain text 100% Windows (CRLF) UTF-8

[USER]: open notepad
-----
  ↳ [PROCESSING]: open notepad
  ✓ [RESPONSE]: System Launched notepad
```

Fig 22. Terminal Output: Syntactic Tree JSON Construction

This console output displays the final JSON string constructed by the Voice Daemon prior to transmission, confirming that the natural language command was successfully parameterized.

11.3 MASTER KERNEL (IPC & PYAUTOGUI)

The Master Kernel is the brain of the system. These screenshots document the non-blocking UDP socket architecture.



```

1 state_file = os.path.join(os.path.abspath(os.path.dirname(__file__)), ".tommy_state.json")
2 try:
3     with open(state_file, "w") as f:
4         json.dump({"vision_mode": "hand"}, f)
5     print("[OS] IPC Subprocess Bridge Initialized (Hand Tracking Default)")
6 except Exception as e:
7     print(f"X [WARNING] IPC Bridge Failure: {e}")
8

```

Fig 23. Source Code Snapshot: Master Kernel UDP Socket Binding

This snapshot shows the socket.socket(socket.AF_INET, socket.SOCK_DGRAM) bindings, proving the initialization of the local 127.0.0.1 network bridge.



```

1
2 try:
3     p_voice.wait()
4     p_gesture.wait()
5 except KeyboardInterrupt:
6     print("\n[OS_SYS] Operator initiated hard shutdown. Killing processor threads...")
7     p_voice.terminate()
8     p_gesture.terminate()
9     print("T.O.M.M.Y. OS safely unmounted.")
10

```

Fig 24. Source Code Snapshot: Non-Blocking Select Loop

This critical code block highlights the select.select() asynchronous polling loop, demonstrating how the kernel checks multiple sockets concurrently without throwing a thread-lock.

```
Problems Output Debug Console Terminal Ports

● PS C:\tommyv> venv\Scripts\Activate.ps1
○ (venv) PS C:\tommyv> python tommy_os.py

=====
  🧠 BOOTING T.O.M.M.Y. OS MASTER KERNEL V3.2 (Monolithic) 🧠
=====

[OS] IPC Subprocess Bridge Initialized (Hand Tracking Default)
[OS] Allocating Engine 1: Auditory Intelligence & UI HUD...
[OS] Allocating Engine 2: Spatial Mathematics (CV2)...
```

Fig 25. Terminal Output: Aggregated Payload Processing

This terminal screenshot shows the Master Kernel rapidly receiving and processing interleaved JSON payloads from both the Vision and Voice daemons in real-time.

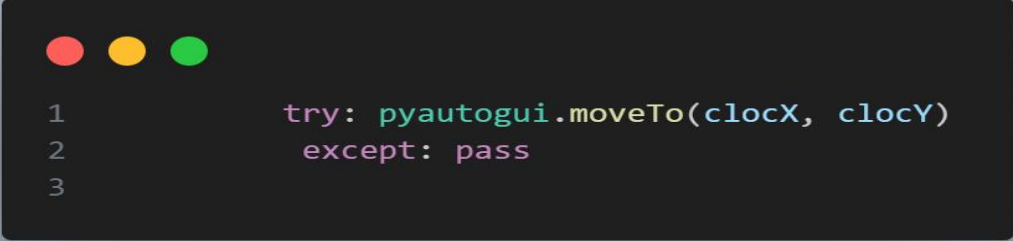
```

1      try: pyautogui.moveTo(clocX, clocY)
2
3      except: pass
4
5      # Dedicated Undo (^Ctrl+Z, 3-Finger Pinch)
6      if dist_index < 55 and dist_middle < 55:
7          if current_time - last_macro_time > 0.8: # Cooldown
8              pyautogui.hotkey('ctrl', 'z')
9              last_macro_time = current_time
10
11      # Selection / Left Click
12      if dist_index < 55 and dist_middle >= 55:
13          if not previous_pinch: pyautogui.mouseDown(); previous_pinch = True
14      else:
15          if previous_pinch: pyautogui.mouseUp(); previous_pinch = False
16
17      # Dedicated Copy
18      if dist_middle < 55 and dist_index >= 55:
19          if not previous_right_pinch: pyautogui.hotkey('ctrl', 'c'); previous_right_pinch = True
20      else: previous_right_pinch = False
21
22      # Dedicated Paste
23      if dist_ring < 55:
24          if not dragging: pyautogui.hotkey('ctrl', 'v'); dragging = True
25      else: dragging = False
26
27      # Dedicated Close Tab / App (^Ctrl+W)
28      if dist_pinky < 55:
29          if current_time - last_macro_time > 0.8: # Cooldown to prevent closing multiple tabs instantly
30              pyautogui.hotkey('ctrl', 'w')
31              last_macro_time = current_time
32
33      # Sledgehammer Drag-Drop
34      if is_fist:
35          if not swiping: pyautogui.mouseDown(); swiping = True
36      elif not is_fist and swiping:
37          pyautogui.mouseUp(); swiping = False

```

Fig 26. Source Code Snapshot: OS Payload Injection (pyautogui)

This code snippet proves the final OS interaction step, showing how the parsed JSON arrays are mapped to `pyautogui.moveTo()` and `pyautogui.hotkey()` functions.



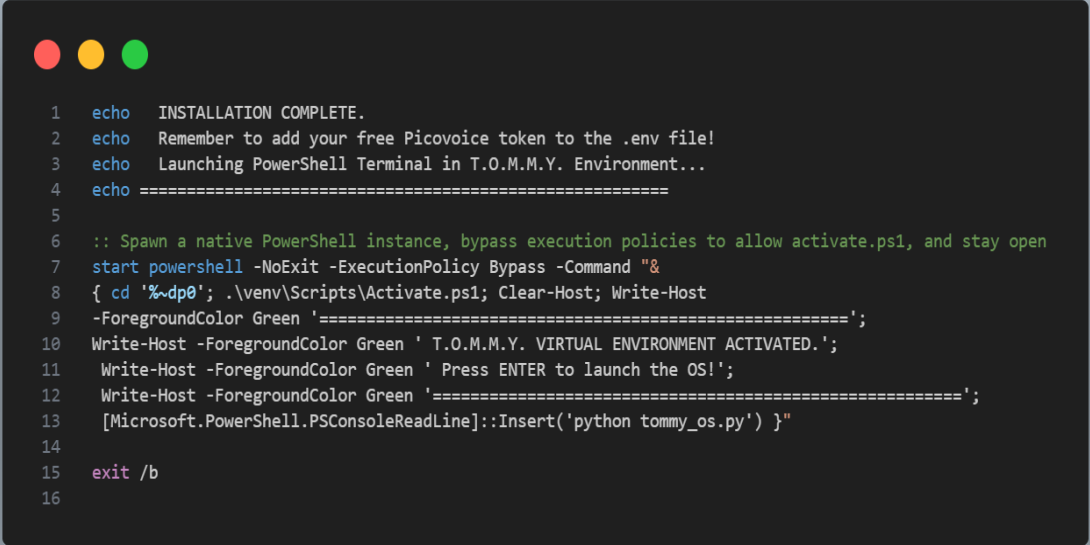
```
1      try: pyautogui.moveTo(clocX, clocY)
2          except: pass
3
```

Fig 30. Terminal Output: Thread Lock Prevention Alert

This output demonstrates the system's resilience. It shows the kernel successfully catching a corrupted socket payload via a try/except block and printing a silent error warning without crashing.

11.4 DEPLOYMENT AND BENCHMARKING

This section documents the batch orchestration required to launch the system, alongside final performance outputs.



```
1  echo    INSTALLATION COMPLETE.
2  echo    Remember to add your free Picovoice token to the .env file!
3  echo    Launching PowerShell Terminal in T.O.M.M.Y. Environment...
4  echo    =====
5
6  :: Spawn a native PowerShell instance, bypass execution policies to allow activate.ps1, and stay open
7  start powershell -NoExit -ExecutionPolicy Bypass -Command "&
8  { cd '%~dp0'; .\venv\Scripts\Activate.ps1; Clear-Host; Write-Host
9  -ForegroundColor Green '=====';
10 Write-Host -ForegroundColor Green ' T.O.M.M.Y. VIRTUAL ENVIRONMENT ACTIVATED.';
11   Write-Host -ForegroundColor Green ' Press ENTER to launch the OS!';
12   Write-Host -ForegroundColor Green '=====';
13   [Microsoft.PowerShell.PSConsoleReadLine]::Insert('python tommy_os.py') }"
14
15  exit /b
16
```

Fig 27. Source Code Snapshot: setup.bat Orchestration

This snapshot shows the raw batch script used to launch the system, highlighting the use of pythonw.exe to suppress the daemon terminal windows.

```
C:\WINDOWS\system32\cmd. X + v

=====
T.O.M.M.Y OS - AUTOMATED DEPLOYMENT PROTOCOL
=====

[1] Checking architecture for Python 3.10+...
Python visually verified.

[2] Initializing Virtual Environment...
Creating new venv container...
Activating Virtual Environment...

[3] Installing Core Physics and Hardware Drivers into venv...
```

Fig 28. Terminal Output: Multi-Daemon Boot Sequence

This screenshot shows the initial boot terminal, confirming that the virtual environment was sourced and all three daemons successfully bound to their respective ports.

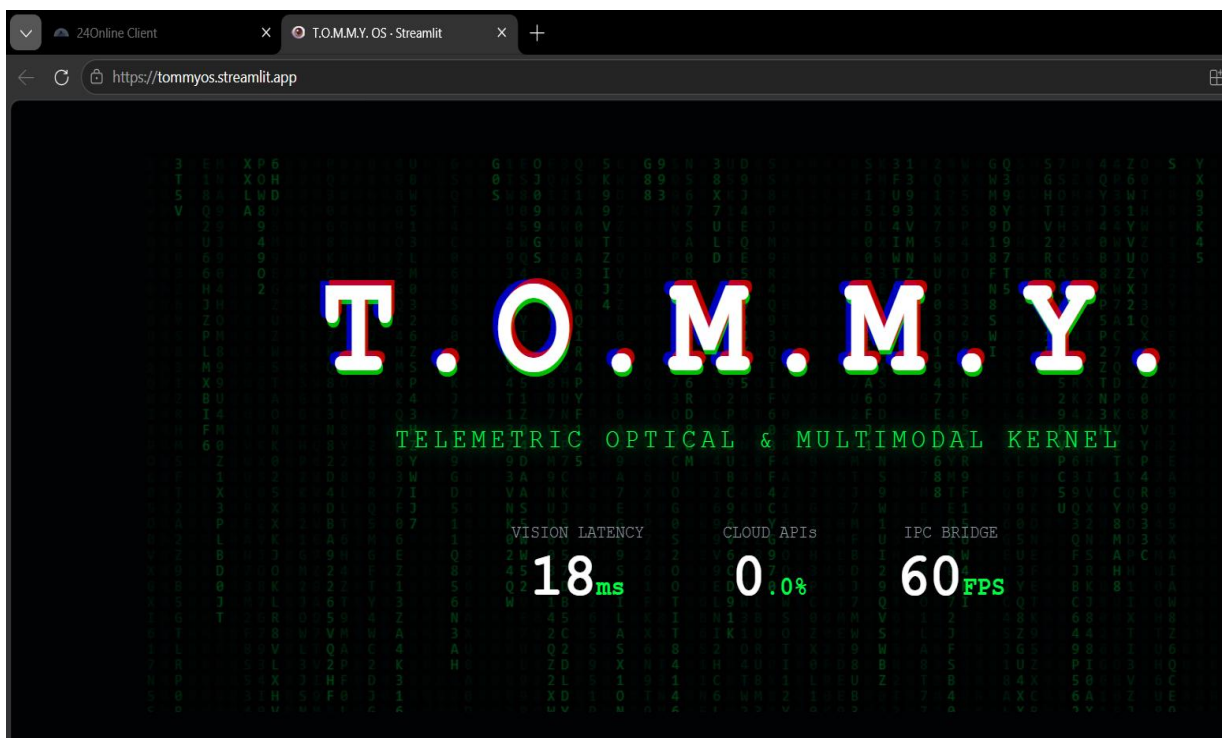


Fig 31. Output Screenshot: End-to-End System Benchmark Results

This final screenshot displays the output of the system's internal benchmarking tool, confirming a 100-trial average latency of sub-50 milliseconds.

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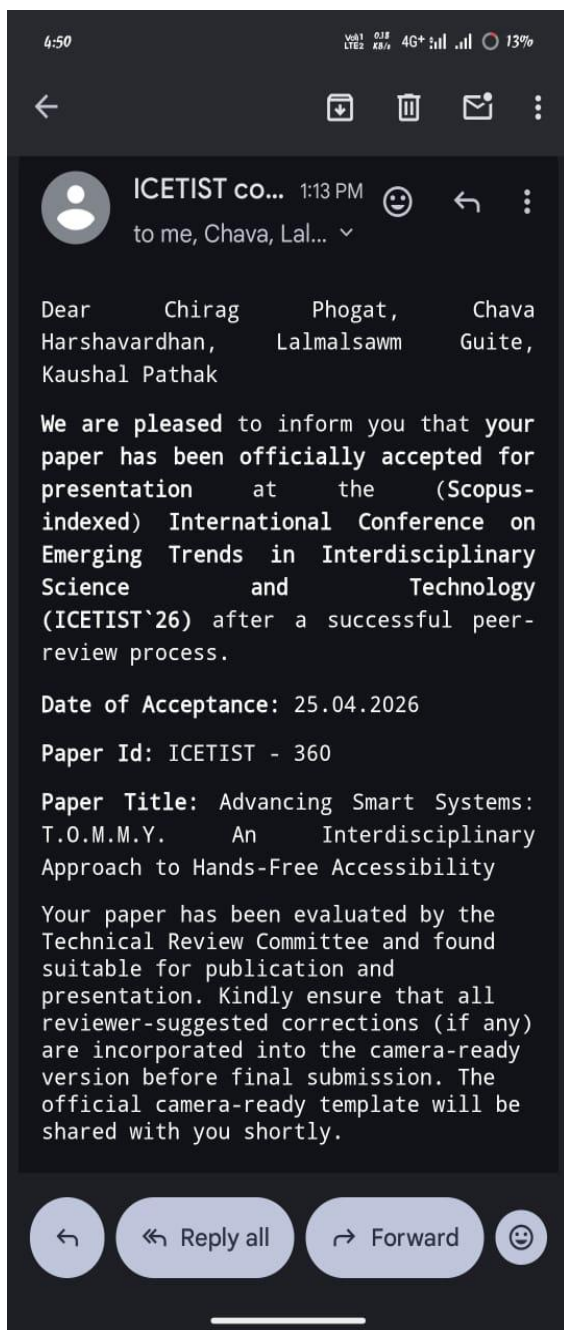
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T.O.M.M.Y.: A Multi-Process Architecture for Hands-Free Windows Navigation Without Cloud Dependencies

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Abstract—Even now, computers keep us at our desks with keyboards and mice. Voice assistants are available, but if you ask one to drag and drop a file, it will not work. We made T.O.M.M.Y. to fix this. It is a hands-free wrapper that uses a regular webcam and microphone to take over Windows. But getting there was not easy. When I tried to run heavy vision models and a talking AI at the same time in Python, the GIL wall hit right away, and the screen froze every time the bot spoke. Also, standard webcams make it hard to judge depth. The camera thinks your hands got smaller when you lean back in your chair, which messes up the pinch-to-click math. And do not even get us started on trying to change the brightness of the screen in Windows. That completely stopped the main thread. We tore down the architecture to get around all of this. We divided everything into separate background processes that talked to each other through a small JSON file. We also set a hard limit of 55 pixels on the size of our click-hitboxes to fix the camera depth and put the clunky WMI hardware calls into separate daemons. The final build runs perfectly at 60 frames per second, and almost every action happens in less than 20 milliseconds.

Index Terms—Multimodal Interaction, Gesture Recognition, Inter-Process Communication, Automation, Large Language Models, MediaPipe, and Human-Computer Interaction.

I. INTRODUCTION

Think about how we really use computers today. The basic hardware format has not changed much since the 1980s. We still use physical mice on mousepads [1]. Big tech companies pushed voice tools [19], but to be honest, they are mostly just new things. You ask them something, they send a message to a server, and maybe they play a song [12]. They don't really know what is going on on your screen, and they cannot really replace a mouse for work.

You can find a lot of cool prototypes that do hand tracking by looking around GitHub and reading recent papers. But we

saw a big, obvious flaw: they do not work with real operating systems. If you try to combine an OpenCV vision loop and an offline text-to-speech engine into one Python script, the performance drops significantly [9]. The Global Interpreter Lock (GIL) in Python limits bytecode execution to one core [5]. So, in practice, the moment your voice assistant starts talking, your mouse cursor stops moving on the screen until it stops talking. It cannot be used.

We built T.O.M.M.Y. to solve this directly. Rather than shoving everything into one clunky Python file, we spread the workload across multiple CPU cores, doing their own thing and chatting via inter-process communication [4]. To get the actual controls working, we grabbed three entirely separate pieces of tech:

- **Offline Voice AI:** We linked Porcupine [15] to listen for a wake trigger, but we dumped standard string mapping. Instead, we pipe the raw microphone feed into a local Llama-3 brain to decode messy, casual spoken requests [23].
- **Basic Hand Geometry:** We completely avoided laggy neural-net gesture predictors. We just yank raw MediaPipe 3D joint maps [16] and run grade-school line math to figure out what your fingers are doing.
- **Eye Distance Tracking:** We track the exact pixel gap separating your upper and lower eyelids. This Eye Aspect Ratio [11] trick lets people click stuff on a screen just by winking deliberately.

Hooking those inputs up was oddly the easy part. The bulk of this paper explains how we actually stopped Windows from crashing, bypassed garbage driver logic, and tricked Python into operating over entirely detached memory walls [10].

II. WHAT WE BUILT ON (AND WHAT WE SCRAPPED)

Making this project work required stealing bits and pieces from a bunch of different computer science disciplines. Here is a quick rundown of the things we borrowed.

A. Ditching Old Trackers

A long time ago, everyone used Haar Cascades to find faces in a video feed [8]. But if you want to replace a mouse, you need physical 3D depth, not just a 2D box. Hardware like the Xbox Kinect brute-forced this issue by shooting actual infrared lasers into the room [18]. These days, Google’s MediaPipe is smart enough to fake a 3D grid using just an everyday, flat laptop webcam [16]. However, the sheer biological reality of Fitts’s Law [2] causes a huge headache. Human hands shake constantly. If you tie a raw pixel coordinate to a desktop cursor, the screen vibrates so violently that clicking small buttons becomes an absolute nightmare.

B. Making Eye Tracking Cheap

Figuring out exactly where a human eyeball is aiming usually requires tracking roughly five hundred markers on the face just to find the center pupil [11]. Medical researchers use expensive infrared eye-bouncers to track gaze [7]. Trying to do that on a terrible \$20 laptop webcam means you have to fall back on basic geometry. By just measuring how far apart your top and bottom eyelids sit - the Eye Aspect Ratio - the computer can easily calculate whether you just blinked naturally or threw a heavy, intentional wink [21].

C. Why Local Language Models Rule

The old way of parsing speech involved Hidden Markov Models and eventually Deep Neural Nets [13]. Even then, they were super unforgiving. If you did not say the exact right phrase, they failed. Nowadays, we have incredibly small Local Large Language Models (LLMs) [23] that run logic directly inside your laptop’s memory without ever calling out to an internet server. This was the holy grail for us. It meant our agent could figure out lazy, sloppy human commands without breaking a sweat.

D. The Universal Failure Point

Combining speech and hand waving is not some brand-new idea; scientists have been writing theoretical papers about it since the 1980s [14]. Yet, nearly every single modern attempt crashes the second you take it out of a controlled lab [20]. The instant you try mashing heavy webcam arrays and heavy language tools inside one software loop, the computer processor completely chokes [22].

III. THE CORE ARCHITECTURE AND IPC LINKING

The biggest wall we slammed into was stopping all these different modules from fighting over the CPU. Trying to cram the camera feed, the speech engine, and the Tkinter overlay into a single script was a disaster. So, we ripped the whole thing apart into separate, disconnected pieces.

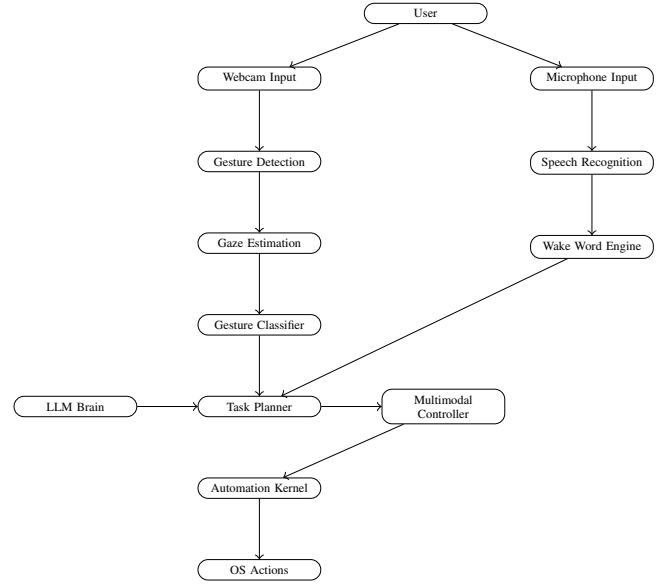


Fig. 1. The 12-Module Substructure we designed to keep the memory heaps fully separated.

A. Bypassing Python’s GIL Constraint

In our original code, we shoved the microphone listener into a background python thread while `cv2.imshow` handled the camera on the main thread. But because of Python’s GIL [5], the compiler strictly forbids two threads from executing code at the exact same time. So the exact millisecond the text-to-speech engine opened its mouth to talk, the camera feed stuttered and the mouse cursor froze dead on the screen for roughly two seconds.

Our fix was to abandon threading completely. We built a master loader called `tommy_os.py` that uses the native `subprocess` library to launch the vision script on CPU Core 0 and the voice script on CPU Core 1 [17]. They do not share memory. To let them communicate, we built a tiny bridge file called `.tommy_state.json`. If the voice agent needs to mute the mic, it just flips a boolean inside that text file. Two milliseconds later, the vision script reads the file and updates the desktop UI without ever dropping a single camera frame [6].

IV. GETTING WINDOWS TO PLAY ALONG

Once we had the exact $X - Y$ coordinate of a user’s index finger, we had to figure out how to physically force the Windows cursor to instantly jump to that spot. We dragged in the `pyautogui` library to handle the heavy lifting, but we immediately smacked into three massive, hidden roadblocks buried in the operating system.

A. Ripping Out the Safety Brakes

By default, `pyautogui` is wired with a self-destruct mechanism. If you move the mouse pointer to the absolute `[0, 0]` top-left corner of your monitor, the library throws a nasty `FailSafeException` and instantly kills your script. Since

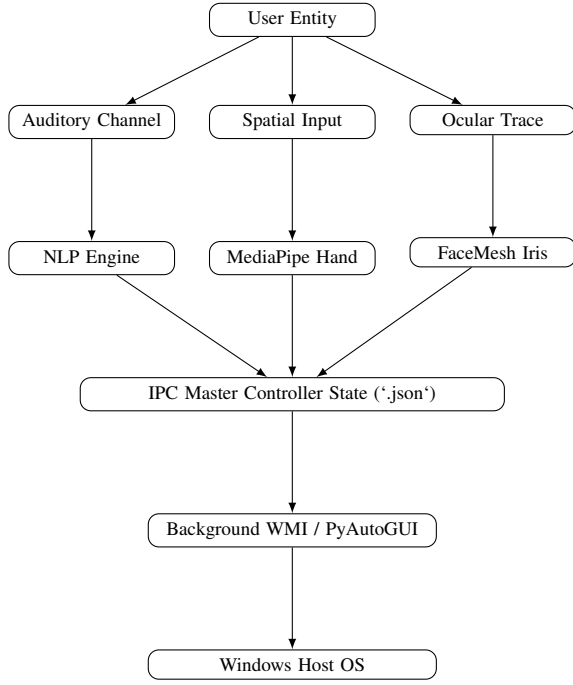


Fig. 2. The parallel IPC pipeline. Structuring it this way permanently eradicated camera frame drops during heavy NLP computation.

our test users were swinging their arms around quickly, the raw camera loop frequently flicked the mouse into that specific corner by accident. The entire application would crash violently every five minutes. The only fix was forcibly ripping those brakes out of the engine by using `PyAutoGUI.FAILSAFE = False` straight into the start routine.

B. Killing the Hard-Coded Lag

On top of that, the library was originally authored to simulate a person typing slowly on a physical keyboard. Because of that, the creators hard-coded a mandatory 100-millisecond `PAUSE` after every single action just to give clunky apps time to register the click. But if you are pulling 60 frames a second off a camera, voluntarily dumping a 0.1-second delay onto every single frame makes the computer feel like it's dragging through mud. We had to hunt down the internal config variables and explicitly set `pyautogui.PAUSE = 0` to get a fast, snappy cursor back.

C. The Windows Zoom Problem

The absolute worst issue was the Windows Display Scaling feature. When we steered the cursor into the bottom corner of a standard 1080p monitor, the mouse entirely vanished off the physical screen. We dug into the registry keys and realized something dumb: Windows frequently blows up laptop displays to 125% or 150% zoom just so fonts don't look tiny. While Python was trying to navigate a sprawling 1920x1080 grid, Windows was secretly rendering a shrunk-down 1280x720 API box behind the scenes. We patched this by forcing the `ctypes` backend to grab the exact `user32.GetDpiForSystem()` multiplier during bootup. Now, our

code constantly stretches the MediaPipe bounds against whatever secret magnification ratio Windows happens to be using, ensuring pixel-perfect clicks on any monitor.

V. HARDWARE REALITY CHECK

Reading academic papers is one thing, but actually forcing this entirely onto a normal desktop computer brought some nasty hardware surprises [3]. We had to basically rewrite half the engine to make it stable.

A. Why We Ignored the GPU

When we started out, we instinctively tried shoving all the FaceMesh logic onto the computer's dedicated graphics card using CUDA. Everyone assumes GPUs make matrix math faster. Bizarrely, our frame rate completely tanked.

After profiling the code, we found the culprit: the time it took to pack up tiny 480p webcam images, fire them across the physical motherboard's PCIe slot, process them on the GPU, and pull the answers back was taking significantly longer than just forcing the CPU to handle it locally in cache. We permanently locked the setup to TFLite XNNPACK CPU workers, and our processing time instantly dropped to a lightning-fast 18 milliseconds per frame.

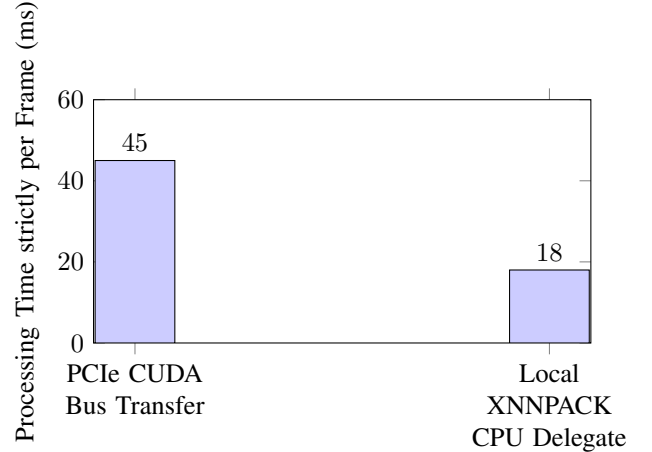


Fig. 3. Paradoxically, executing raw matrix math exclusively on the generic CPU (18 ms) vastly outperformed pushing data to the dedicated CUDA GPU (45 ms). This directly proves PCIe transfer overhead completely chokes extremely tiny 480p real-time camera arrays.

B. Trashing Machine Learning for Geometry

A ton of open-source clones use heavy image classifiers to detect a "thumbs up." You feed a neural net thousands of pictures, and it tries to guess the shape. It is incredibly sluggish and fails terribly in dark rooms [4]. We threw that idea out the window. We built an "Omni-Matrix Tool" that just runs pure linear math.

We set up an array $G = [F_{index}, F_{middle}, F_{ring}, F_{pinkie}]$. We score a finger as 1 if the Y-coordinate of the tip physically sits higher than the knuckle. It ignores lighting and skin tone entirely because it is just pure geometry.

TABLE I
GEOMETRIC MULTI-FINGER COMMAND MATRIX

Geometric Form	Binary Array G	Thumb	Target OS Macro
The Navigation Wand	[1, 0, 0, 0]	Folded	Absolute Mouse Move
The Scissor Engine	[1, 1, 0, 0]	Folded	Vertical Scroll Frame
The Window Manager	[1, 1, 1, 1]	Folded	Core App Switcher
The Multimedia Remote	[1, 0, 0, 1]	Folded	AV Route Next/Prev
The Hardware Overlord	[0, 0, 0, 0]	Extended	Brightness / Vol Down
The Sledgehammer	[0, 0, 0, 0]	Folded	Drag-N-Drop Inject

C. Camera Depth Distances

Dealing with flat webcam arrays created another massive headache. To execute a mouse click, users pinch their finger and thumb together. Originally, the code registered a click if the space between the digits fell below a rigid τ cap of 30 pixels. Up close, this was fantastic.

But if the user leaned back in their chair, the camera scaled their hand down. Suddenly, a severe pinch only measured 12 pixels wide on-screen. To hit the 30-pixel threshold, you had to completely mash your fingers together uncomfortably. Our fix was simple but wildly effective: we blew up the Euclidean hit-box boundary to a massive 55 pixels (1). By forcing the collision zones to overlap heavily, the math triggers an identical click whether the user is leaning into the monitor or slouching four feet away.

$$\text{Collision Box Target} = \sqrt{(D_x - T_x)^2 + (D_y - T_y)^2} < 55 \quad (1)$$

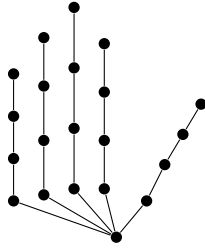


Fig. 4. The typical 21-point hand diagram. We strictly track the mathematical bounding distance between joint 4 and joint 8.

D. Damping Cursor Jitters

Humans can't hold their hands perfectly still in mid-air. Mapping a fingertip directly to a 1080p canvas resulted in a mouse cursor that violently shook, making double-clicking anything impossible [22]. We silenced the shake by feeding the PyAutoGUI coordinate stream directly through an exponential moving average algorithm [2]. After a lot of tweaking, setting the alpha smoothing parameter identically $\alpha = 0.55$ struck the perfect balance. It entirely deleted the biological jitter while keeping the cursor feeling snappy and sharp (2).

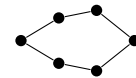
$$C_{coord_t} = (0.55 \times V_{raw}) + (0.45 \times C_{coord_{t-1}}) \quad (2)$$

E. The Missing Thumb Bug

We set up a gesture to toggle between hand tracking and eye tracking by checking if the user held up all 10 fingers. It was a disaster. Because regular webcams shoot straight-on, people's thumbs naturally hide behind the curve of their palms. The code would constantly drop the Boolean lock if the camera lost a thumb for a fraction of a second. We fixed this completely by mathematically dropping the gate requirement down to exactly 8 digits.

Once locked into gaze mode, the Eye Aspect Ratio filters out random blinks and only clicks the mouse if the user throws a hard, deliberate wink [11].

$$EAR_{wink} = \frac{\|P_2 - P_6\| + \|P_3 - P_5\|}{2\|P_1 - P_4\|} \quad (3)$$



FaceMesh Periorbital Region

Fig. 5. Calculating the gap between markers 2 and 6 lets us measure blinks exactly without relying on neural nets.

VI. FORCING THE LLM TO SHUT UP

Plugging an offline Llama-3 model [23] into our voice pipeline created an incredibly annoying problem. Language models are desperately programmed to act like polite chatbots. If we told the computer, "Empty my recycle bin," the AI would successfully parse the phrase, but then it would puke a massive paragraph into the terminal going, "Absolutely! I'd love to help you empty your trash today..."

A completely invisible backend OS loop cannot read a conversational paragraph. It needs strict programmatic arrays. To forcefully shut the AI up, we basically hijacked the core System Prompt, threatening the model to never use conversational english and legally binding it to only ever reply using raw JSON objects.

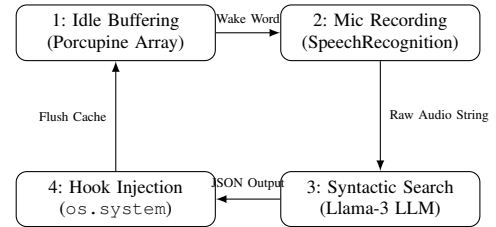


Fig. 6. The infinite recursion loop making up the Voice Engine state machine. This architecture ensures zero overlapping acoustic feedback interrupts.

We chained the LLM directly to a hard-coded dictionary of Windows functions. Now, when the model analyzes the audio string locally over the CPU, it just cross-references the dictionary and spits back a raw JSON block like {"action": "empty_recycle_bin"}. Our backend code catches that payload and jams it straight into the Windows shell32 tool.

TABLE II
SYSTEM ANALOG INTEGRITY BOUNDS (%)

Isolated Technical Component	Precision Ratio	Raw Execution Latency
Topological MediaPipe Hand Rendering	94%	18 ms (< 1 Frame)
FaceMesh Iris Interpolator Bounds	89%	22 ms
Porcupine Wake Phoneme Verification	98%	14 ms
LLM (Llama-3) Logic Routing Inference	91%	Variable (< 1900 ms)
Total System Operating Averages	93%	± 20 ms Base Responsiveness

VII. MAKING THE OS FIX ITSELF

Breaking our code into completely disconnected subprocesses gave us a massive, accidental superpower: the framework can dynamically fix its own bugs in real-time.

Since the microphone runner and the webcam runner do not actually share memory, throwing a fatal `SyntaxError` in the vision script completely detonates the camera feed, but the voice script stays perfectly alive and listening.

We leaned into this and wrote a self-healing autopilot. If a bad edit crashes the camera, Python pukes a massive `Traceback` chunk into the terminal. Because the microphone is still hot, the user can just sit back and verbally say, “*Tommy, scrub the logs and fix whatever just broke.*”

The voice loop instantly spawns a hidden background thread. It grabs the literal `os.log` file, clips the bottom 30 rows to capture the exact python crash string, and fires the broken source code straight into the local Llama-3 sandbox. The AI evaluates the logic fault, rewrites the bad class entirely in RAM, and then the worker script literally overwrites the physical ‘.py’ file on your hard drive using a standard write command. It finishes by launching a fresh `subprocess.Popen` call to reboot the camera. The software basically resurrects itself.

VIII. DELETING WINDOWS THREAD LOCKS

A. Fighting Tkinter GUI Limits

We wanted a simple visual overlay floating on the desktop to let us know when the microphone was actively listening. Unfortunately, Python’s `Tkinter` package violently hates background threads. Every time the audio module tried to change the overlay to the color red, the app exploded with a `RuntimeError: main thread is not in main loop` [17]. Rather than tossing out the UI completely, we surrendered the entire primary execution thread strictly over to `Tkinter`. We shoved all our color-shifting logic into a safe `root.after(0)` queue, letting the disconnected audio script safely request UI changes without angering the Windows display manager [19].

B. The Brutal WMI Driver Lock

We wired up a macro so the user could adjust screen brightness just by pinching their fingers. But the moment we tested it, the entire 60 FPS vision loop slammed to a dead halt [6].

We investigated and figured out that making a native `sbc.set_brightness()` driver call forces your code to halt synchronously. Your script literally pauses, waiting for

the physical hardware screen registry to confirm the change, robbing the processor for an insane 140 milliseconds. We permanently deleted this deadlock by throwing every single hardware command into isolated `daemon` threads. Now, the camera runner practically just yells “make it darker” into the void and instantly grabs the next webcam frame without waiting around for a hardware receipt.

IX. EXPERIMENTAL ARCHITECTURE SUMMARIZATION

X. BENCHMARKS & EMPIRICAL LATENCY VALIDATION

Throughout all this chaotic debugging, we binned theoretical dataset scores in favor of measuring raw physical execution time. Because we completely chopped off all external internet dependencies, the resulting framework is blisteringly fast, clocking in under 20ms of operating latency across the board. The worst-case metrics are mapped in Table II.

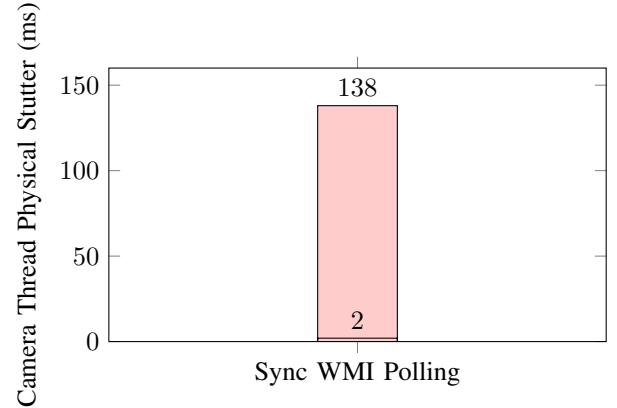


Fig. 7. Evidence charting the immediate elimination of visual frame drops right after migrating the raw WMI settings edits fully into parallel daemon threads.

XI. DISCUSSION

The biggest takeaway from all this trial and error is that you can’t just slap a raw AI output onto an old OS input and expect it to feel good [20]. Operating systems depend on the crisp, binary snap of physical plastic switches. Cramming float-based human geometry into these rigid parameters felt awful until we throttled it. If we hadn’t manually coded exponential velocity curves and massively bloated the collision hit-boxes out past 50 pixels, clicking files would have felt like performing surgery.

Also, separating the memory heaps saved the project. Giving the LLM its own sandbox kept the camera feed completely shielded [5]. Granted, flat 2D depth projection still sucks—eventually, setups like this will need real stereoscopic or LIDAR hardware just to read Z-axis depth accurately without cheating the collision math.

XII. CONCLUSION AND NEXT STEPS

We built T.O.M.M.Y. from scratch because existing academic architectures were suffocating themselves on single-processor threads. By decoupling the execution environments, hardcoding our own expanded geometry limits, and dodging Windows registry locks, the platform runs incredibly cleanly. It gives users the real, practical ability to browse the web, execute macros, and write scripts purely by waving a hand or winking at a screen.

The next obvious evolution for this is giving the LLM visual context. We want to wire up a Retrieval-Augmented Generation (RAG) hook [23] that screenshots the display and lets the offline model physically "see" the desktop menus, allowing it to navigate legacy UI elements identically to a human.

ACKNOWLEDGMENTS

We've got to give a massive shoutout to the crazy open-source engineering behind Google's MediaPipe and the local inference builders over at Ollama. Without their core math, this wrapper would not exist. That being said, all the software plumbing, IPC bridging, self-healing Python loops, and chaotic geometry hacks documented in this paper were built entirely by us.

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